

ARCHITECTURES OF INTERACTIVITY: MOVEMENT, SYSTEMS, AND THE BLACK BOX

by Polina Rud



1. FROM OBJECT TO FIELD: THE EMERGENCE OF ACTIVATION

- Why activate the viewer?
- What changes when perception becomes action?
- Who gains power when the spectator becomes a participant?
- Is participation the same as freedom?
- What kind of subject is being trained through interaction?

2. CYBERNETICS FEEDBACK AND THE BLACK BOX

- The invention of feedback
- Distributed control and homeostasis
- Systems that respond and adapt
- Communication as regulation
- Participation inside coded limits
- Adaptive machines and behavioral design
- The emergence of the black box

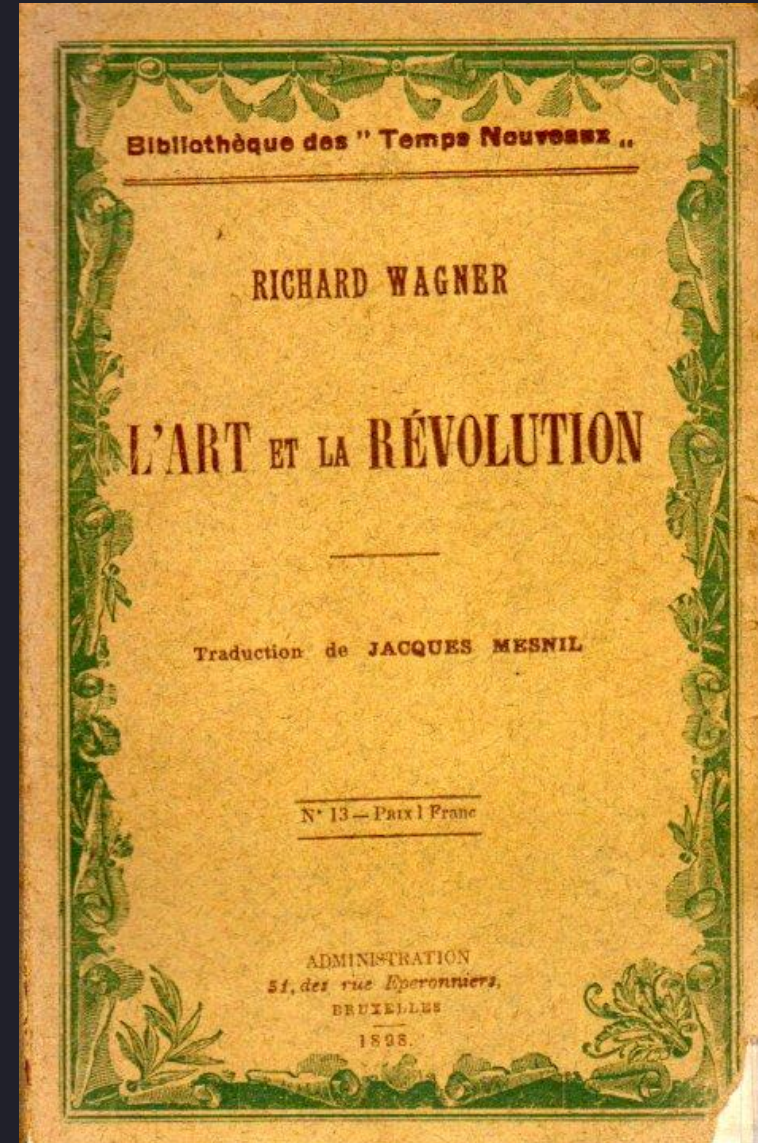


FROM OBJECT TO FIELD:
THE EMERGENCE OF ACTIVATION

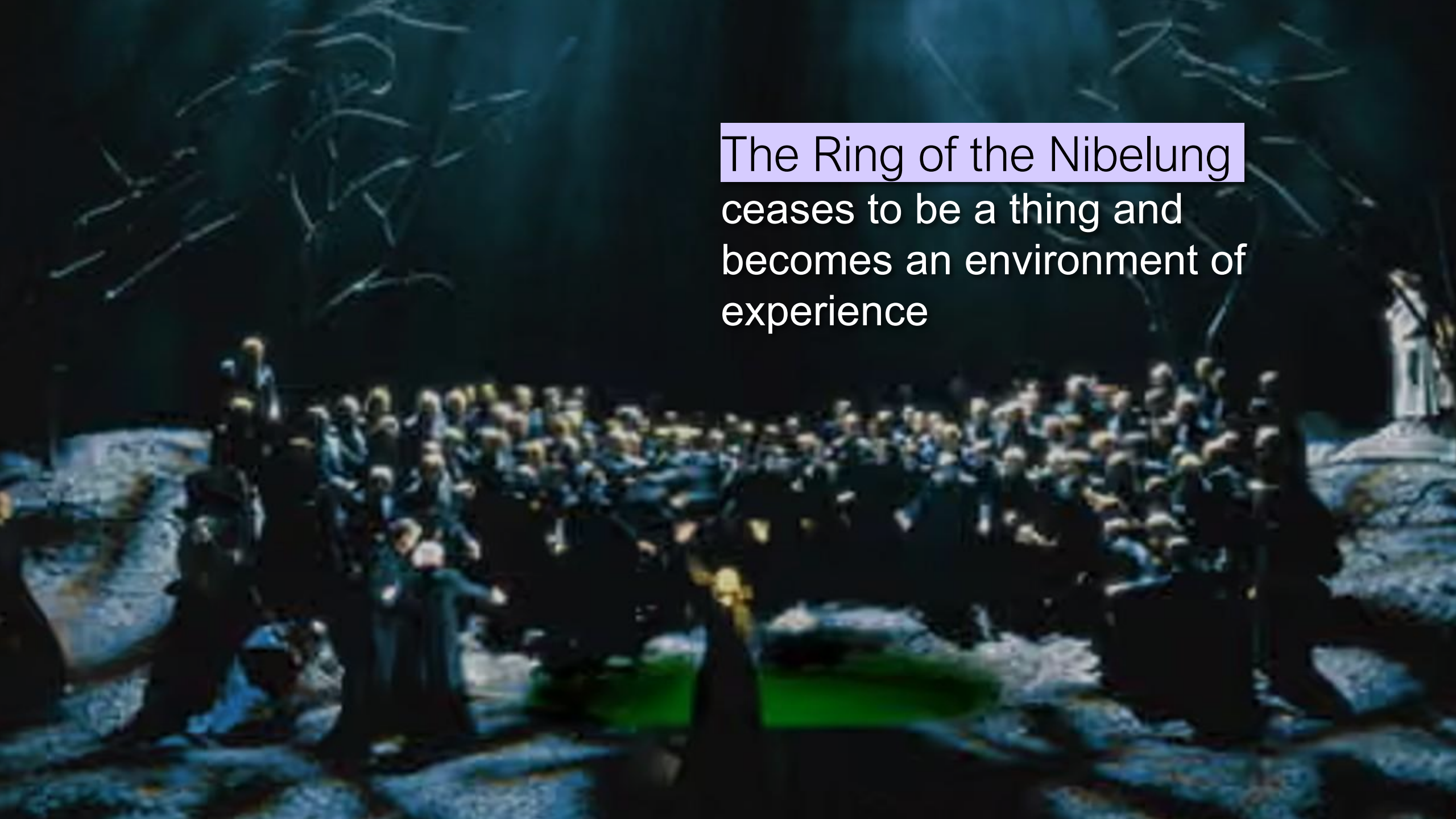
The object is static.

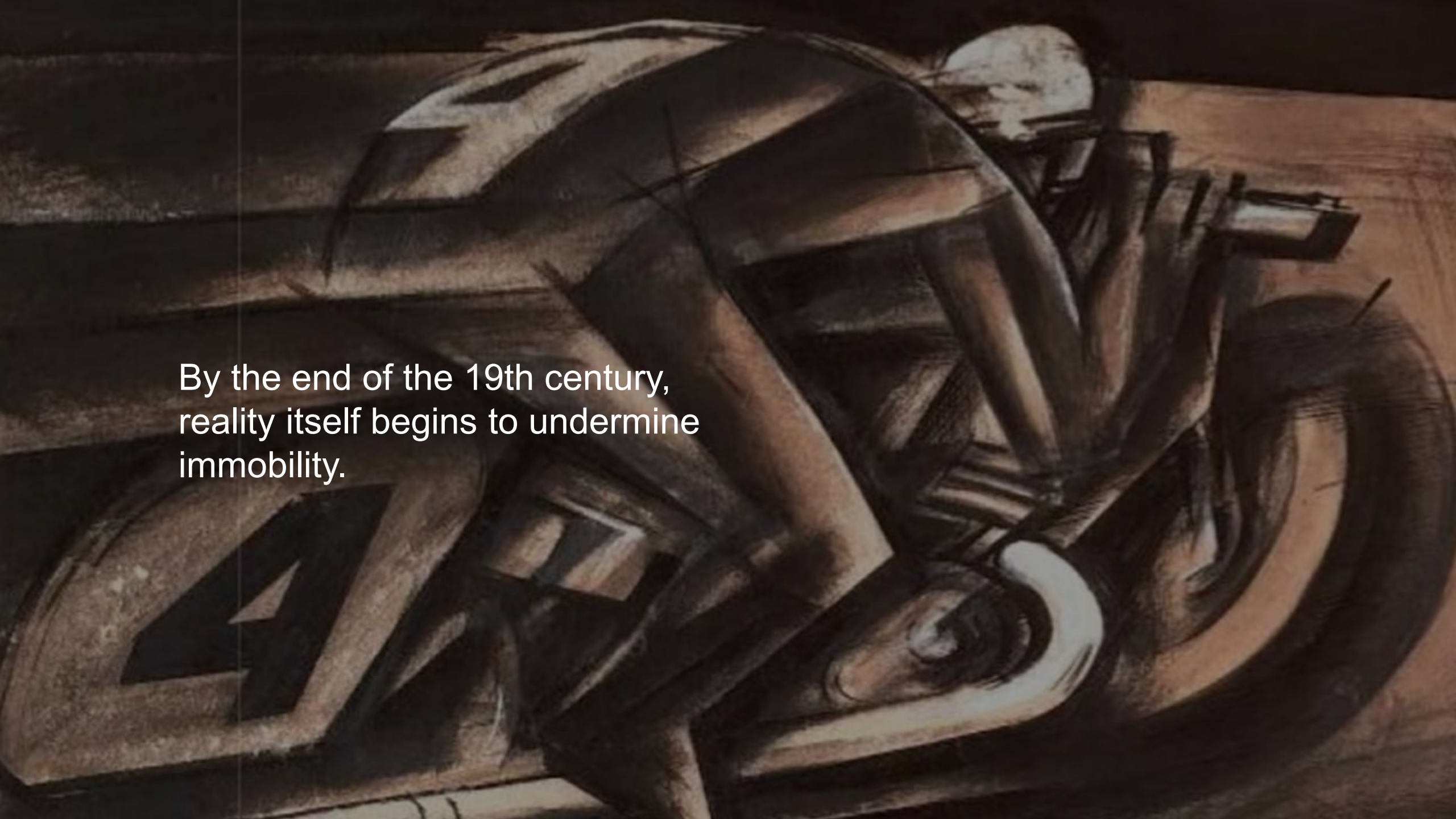
The viewer is an external observer.

Gesamtkunstwerk is meant
for collective catharsis

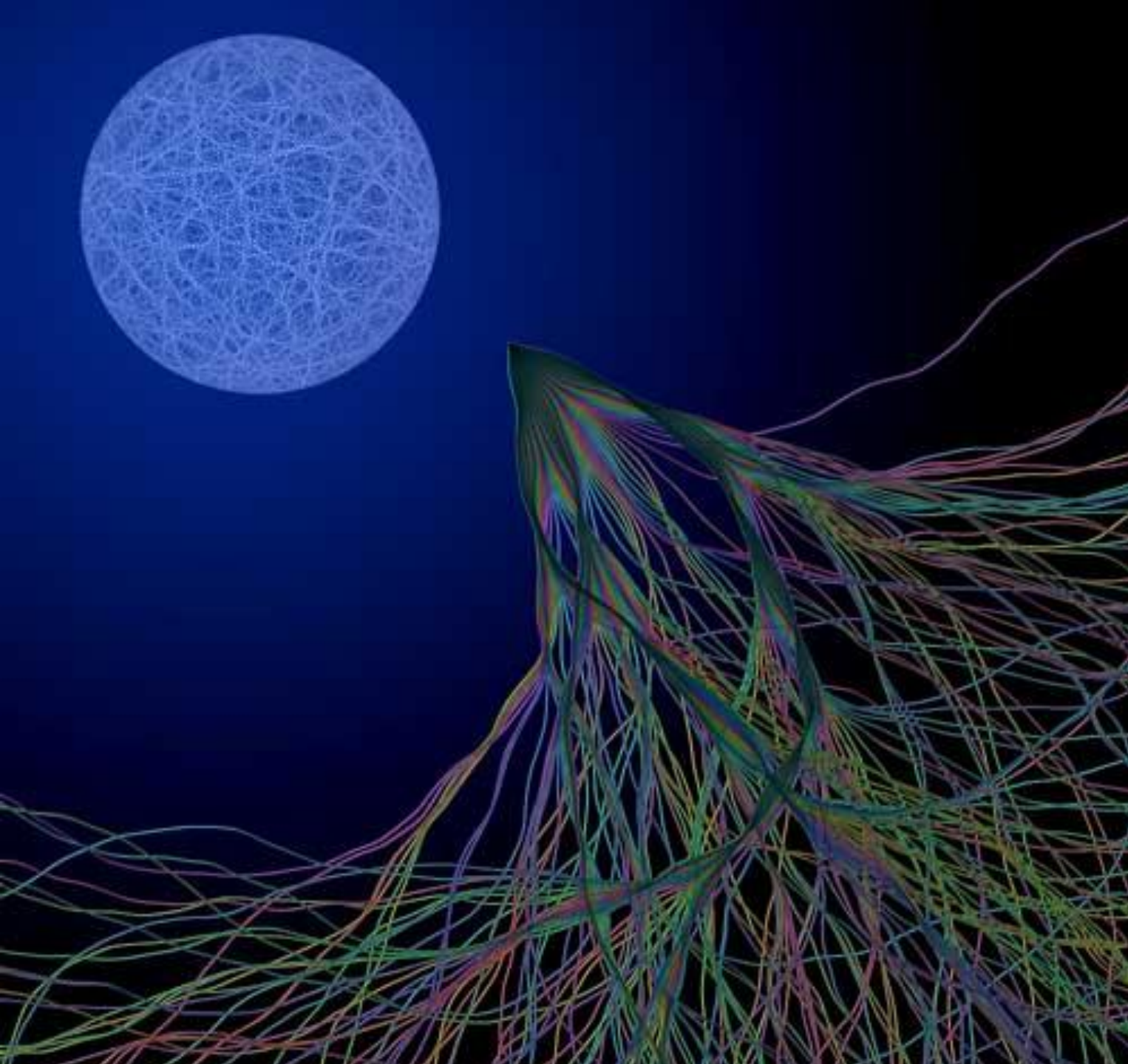


The Ring of the Nibelung
ceases to be a thing and
becomes an environment of
experience



The background is a dark, abstract painting. It features a large, stylized number '4' in the lower-left quadrant. The painting is composed of various geometric shapes, including triangles, rectangles, and curved lines, rendered in shades of brown, black, and white. The overall style is reminiscent of Cubism or Abstract Expressionism. The text is overlaid on the left side of the image.

By the end of the 19th century,
reality itself begins to undermine
immobility.

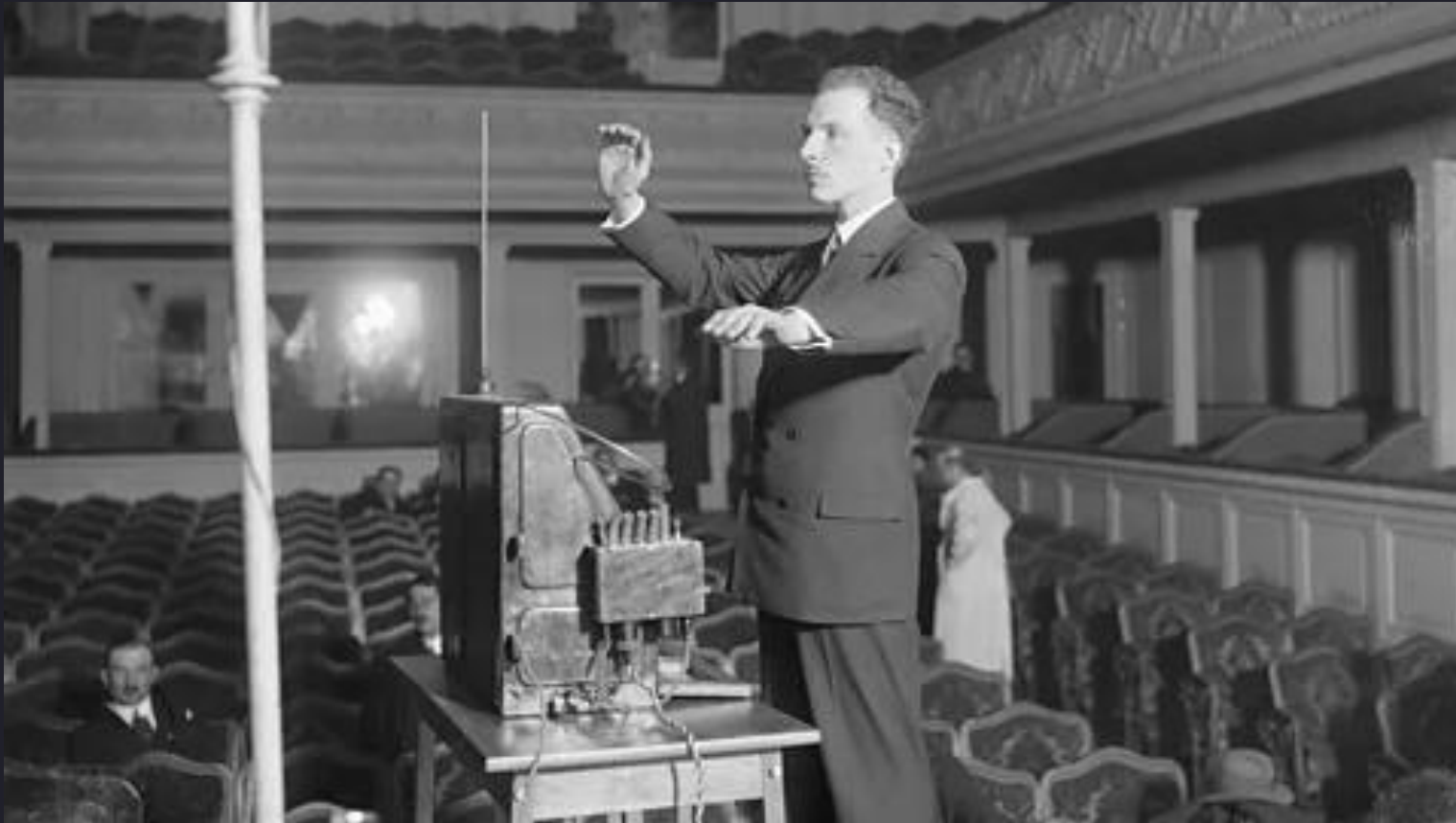


Matter is no longer mass. It is process. If reality is process, then art that wants to be modern cannot remain static.



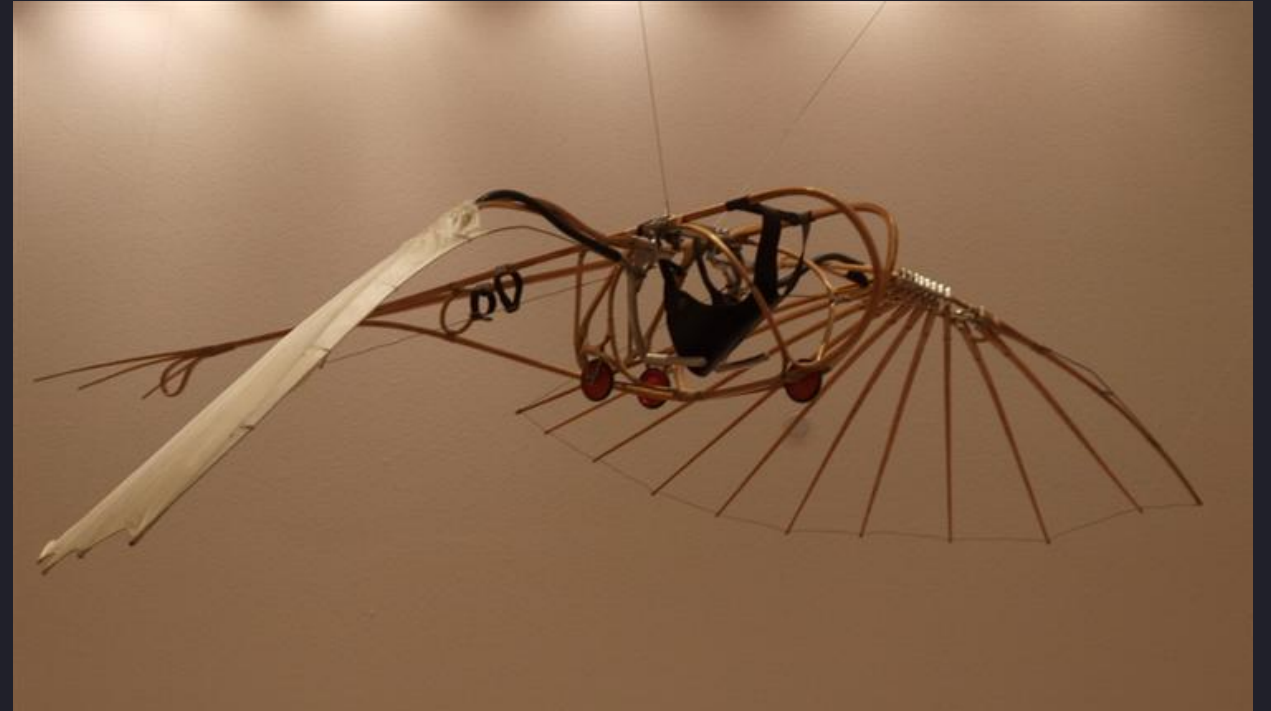
Marcel Duchamp, Bicycle Wheel (1913) and Roto-Reliefs (1935)

the viewer triggers the object



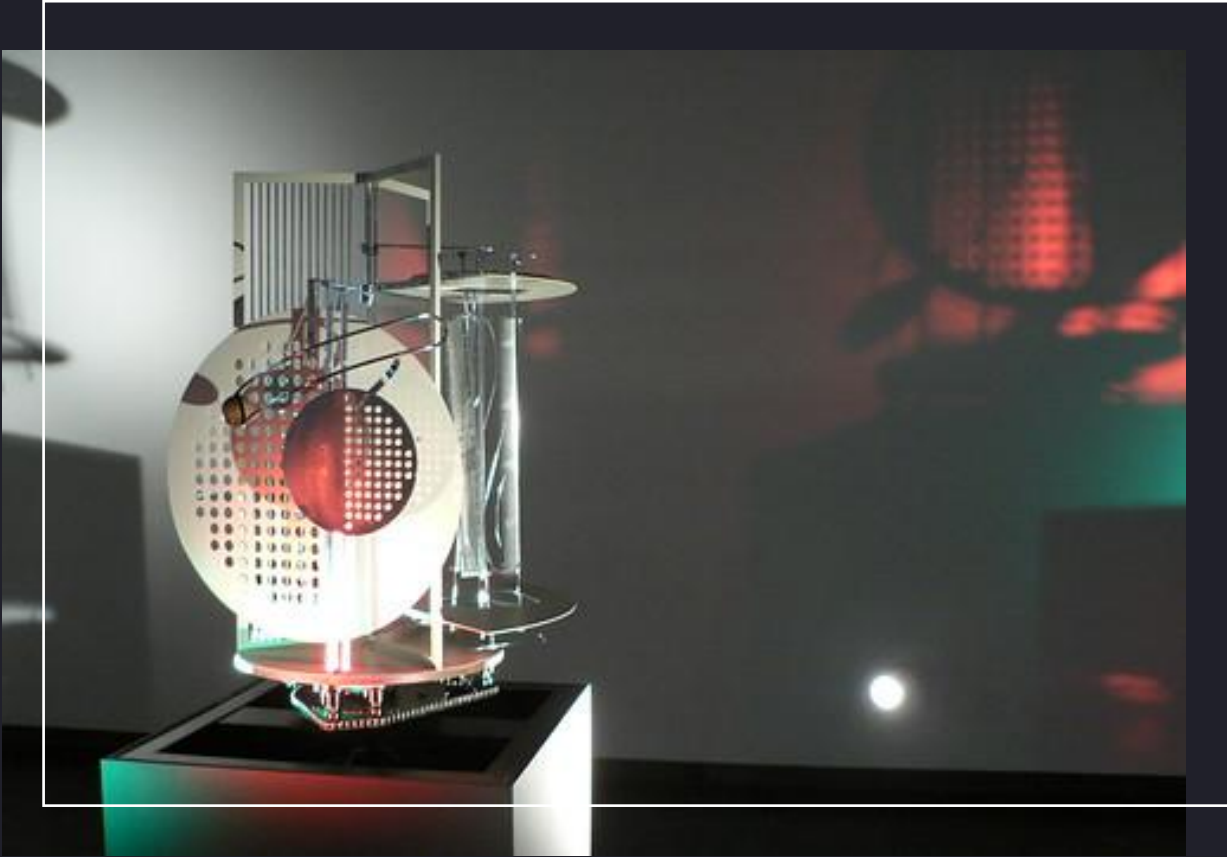
Leon Theremin and his theremin (1920)

The body enters an invisible field



Vladimir Tatlin, Letatlin (1929–1932)

the body becomes part of the machine itself



László Moholy-Nagy, Light-Space Modulator



The artwork is a field of forces,
and the spectator is an active
factor within it.

Moholy-Nagy and Alfred Kemény in
Manifesto of Dynamic-Constructive Energy

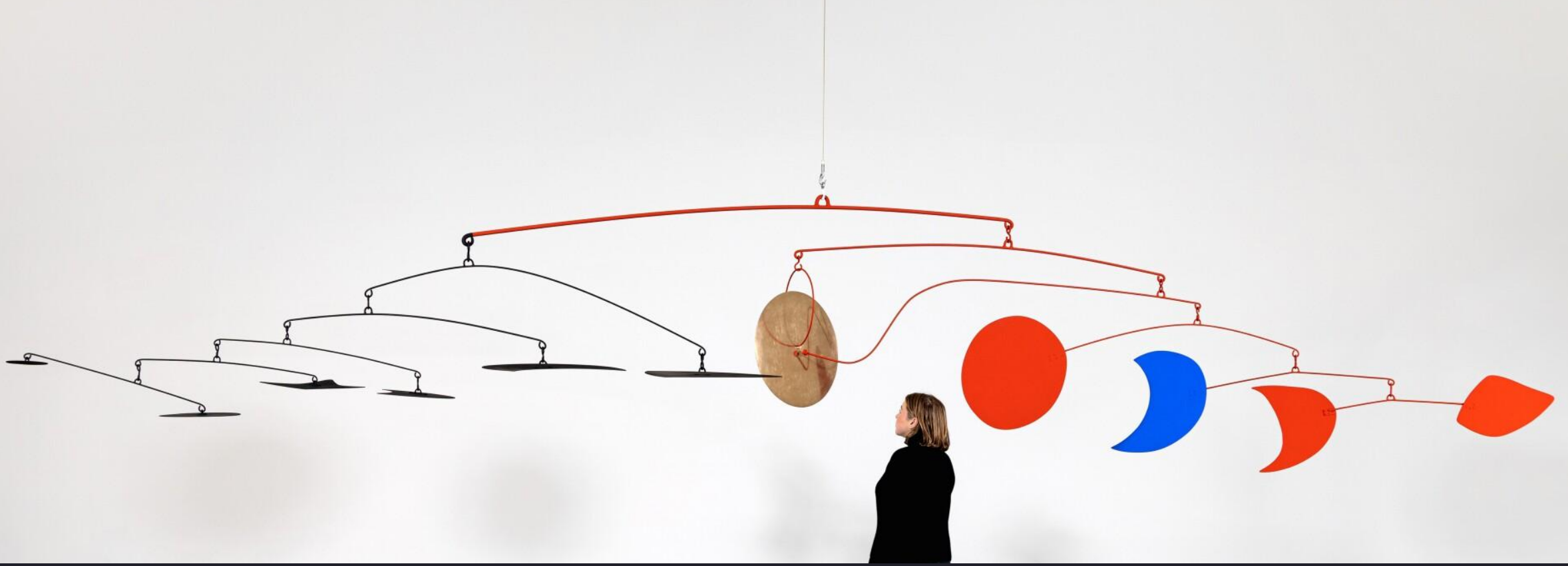


Sculpture no longer
represents movement.
It is movement.

Interactivity through
perception.

Kinetic Construction
(Standing Wave), Naum Gabo



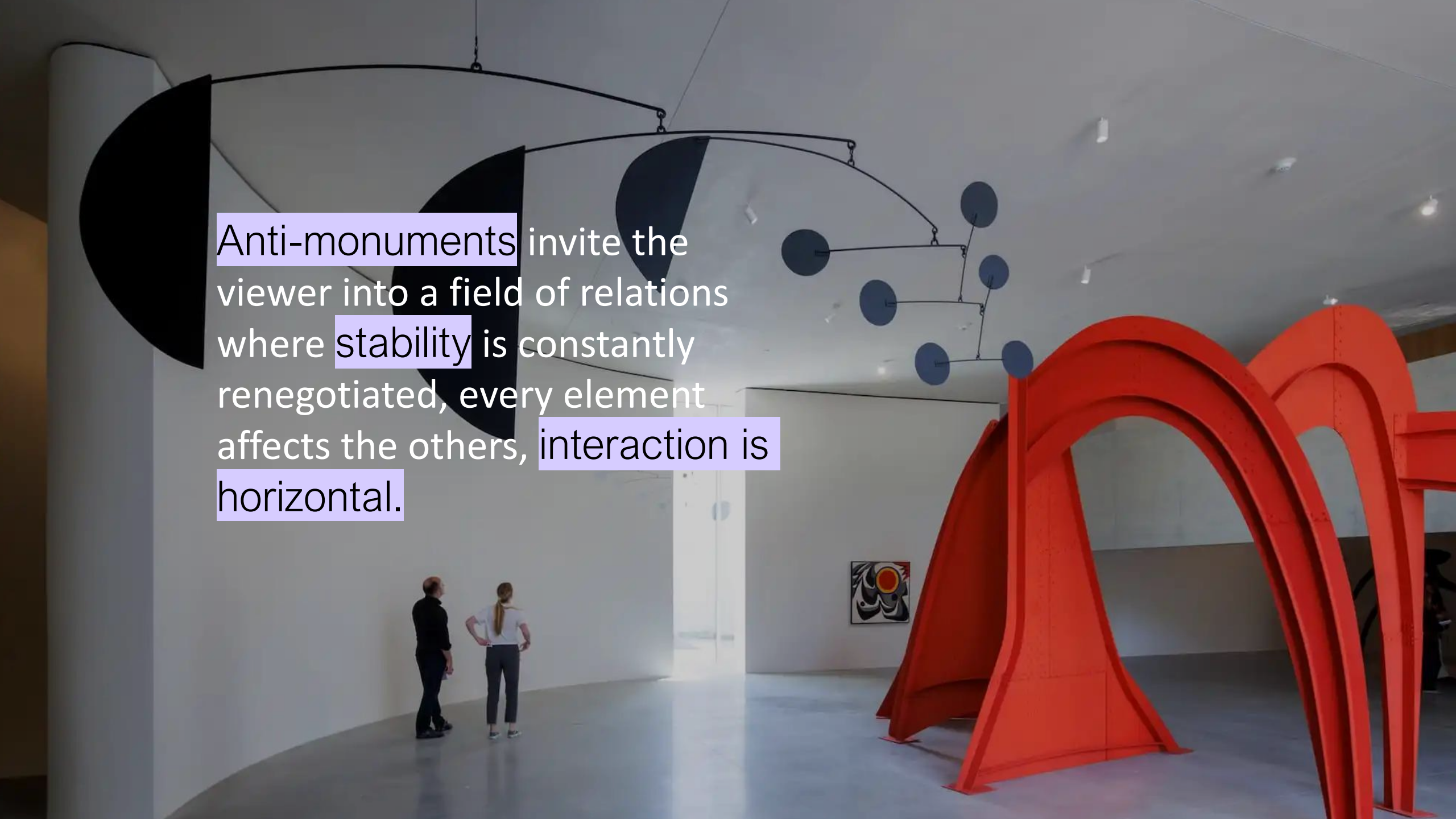


Alexander Calder's mobile

The movement emerges through interaction with the air.
Your body moves air, and the sculpture responds.
Interactivity happens through a shared field.

There is no strict
hierarchy between artist
and viewer.

Anti-monuments invite the viewer into a field of relations where stability is constantly renegotiated, every element affects the others, interaction is horizontal.



When you want to communicate an idea through your artwork, where does that communication actually happen?

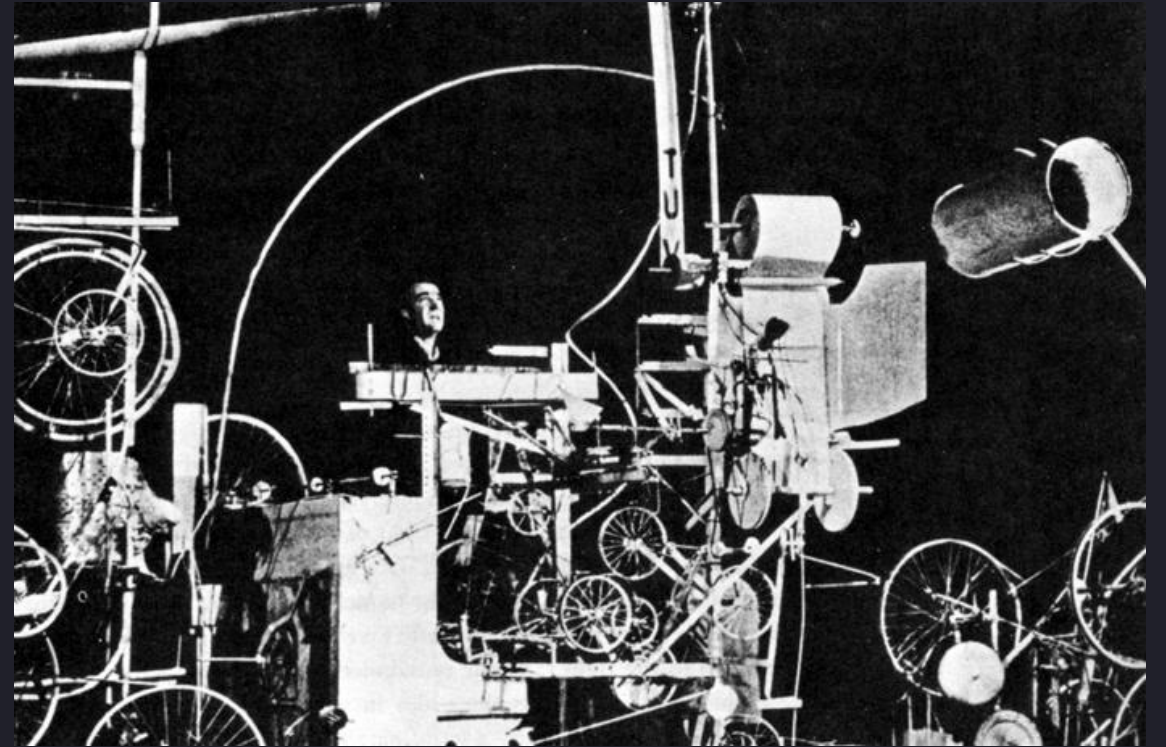
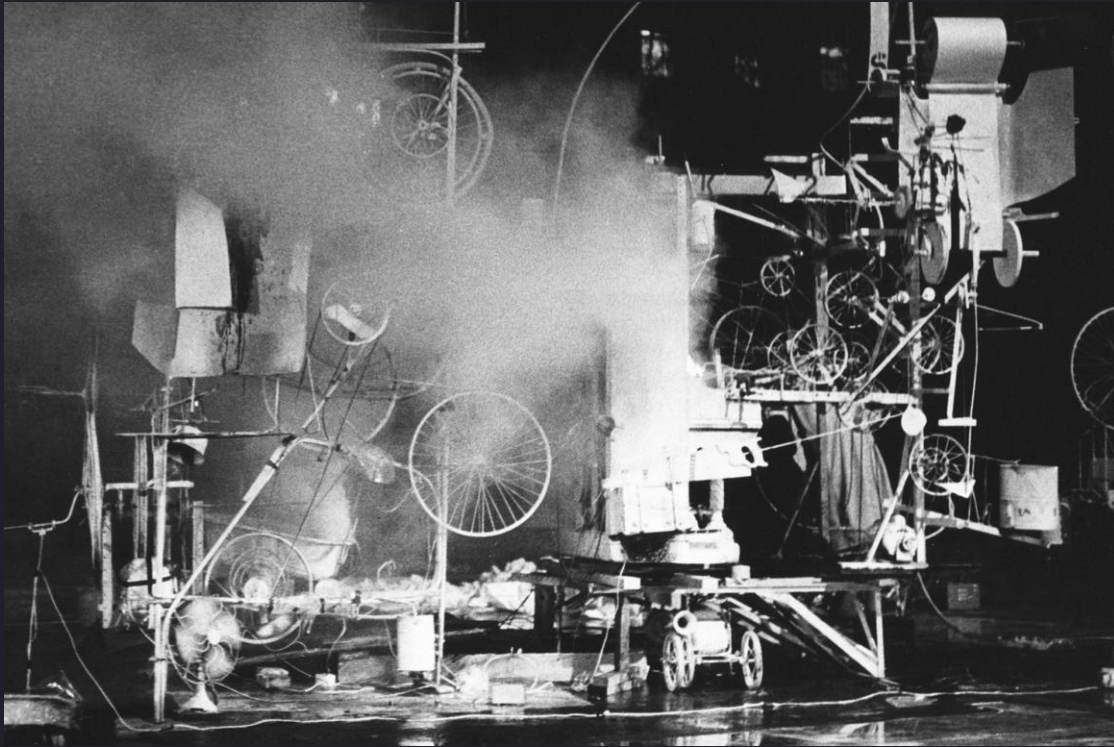
Does it emerge mainly through what you say about the work (the concept, the text, the narrative, the images) or does it already exist within the form itself?

If everything else were removed and only the form remained (the material, structure, movement, scale, balance, rhythm), what would your work communicate?

And if you had to express that communication through one single word conveyed only by form, what would it be?

Studio Reflections





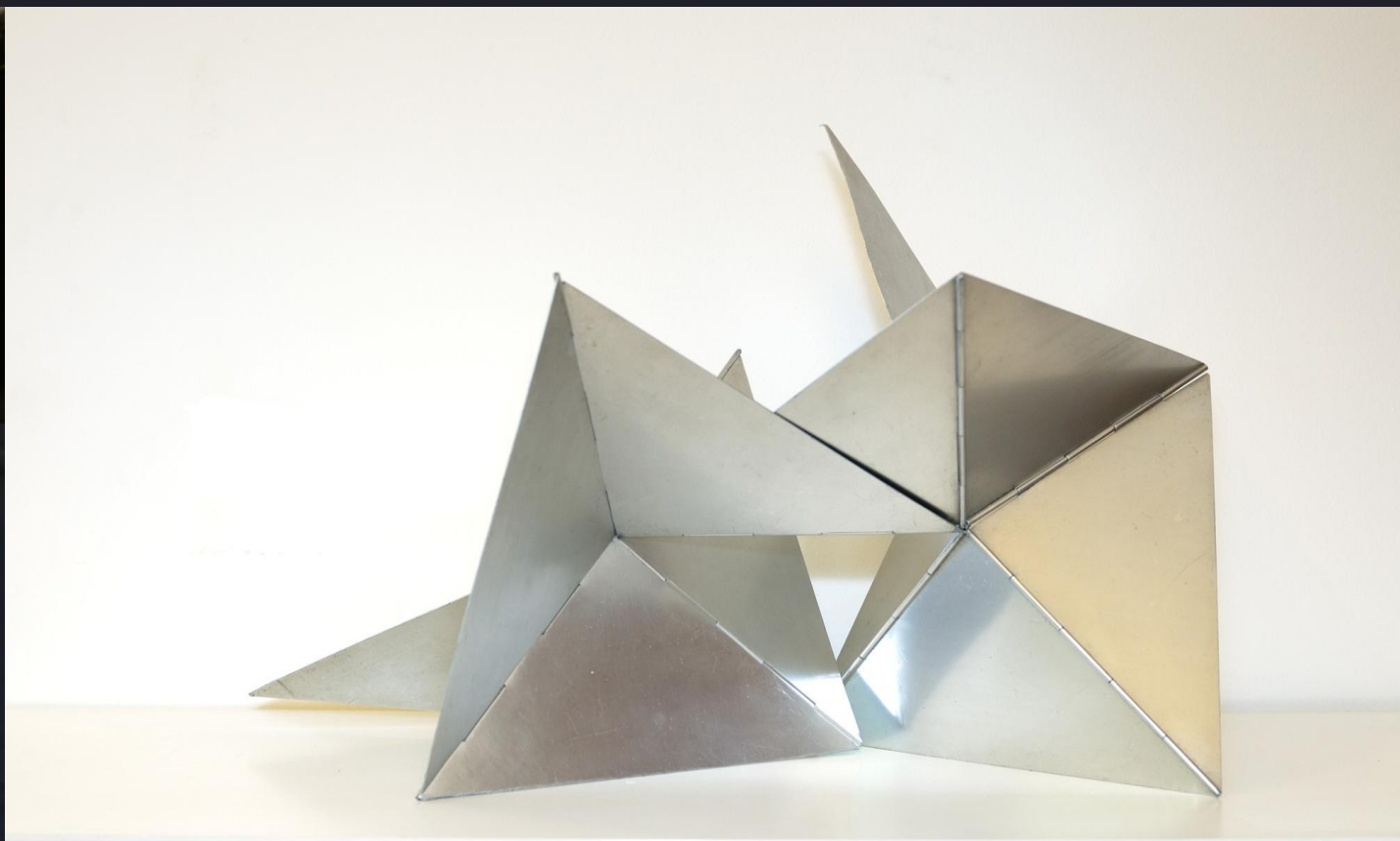
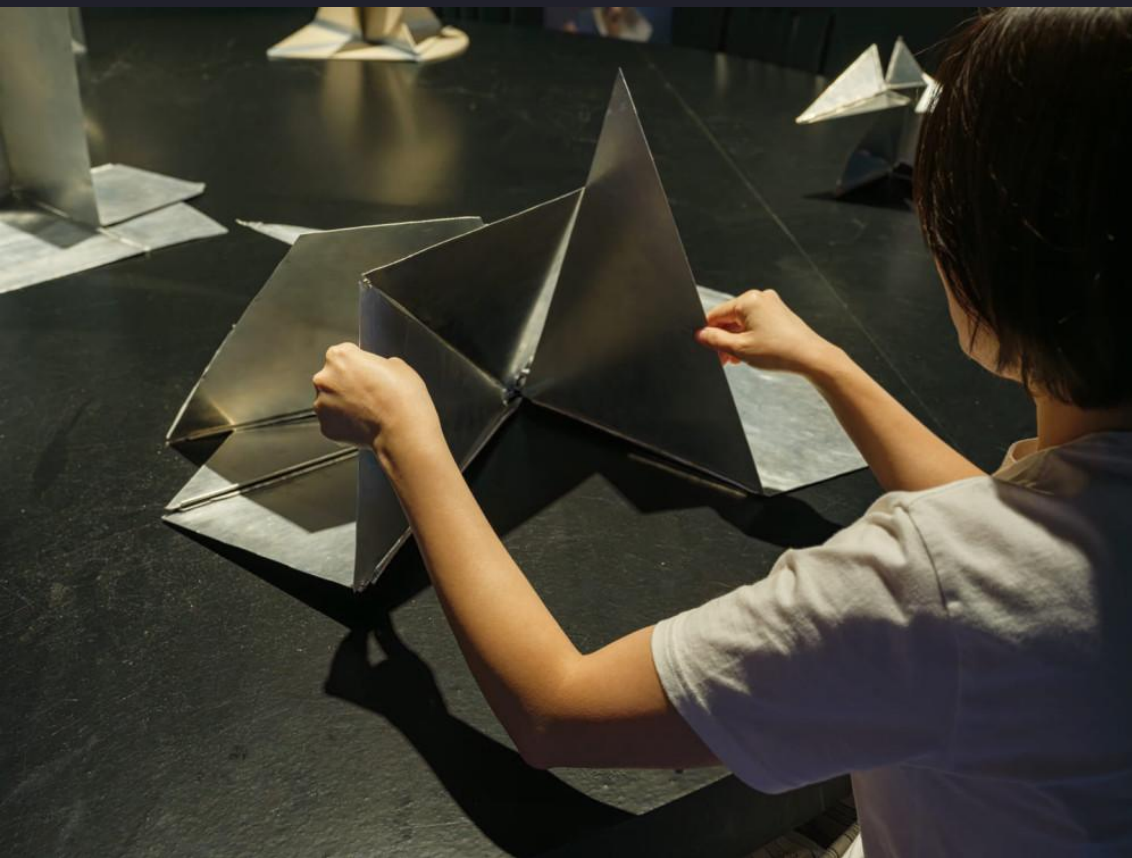
Jean Tinguely, *Homage to New York* (1960)

In his “metamechanics,” the machine does not gently respond to presence. The viewer must adapt to the system’s behavior.



Yaacov Agam, tableaux transformables (1953)

In his “metamechanics,” the machine does not gently respond to presence. The viewer must adapt to the system’s behavior.

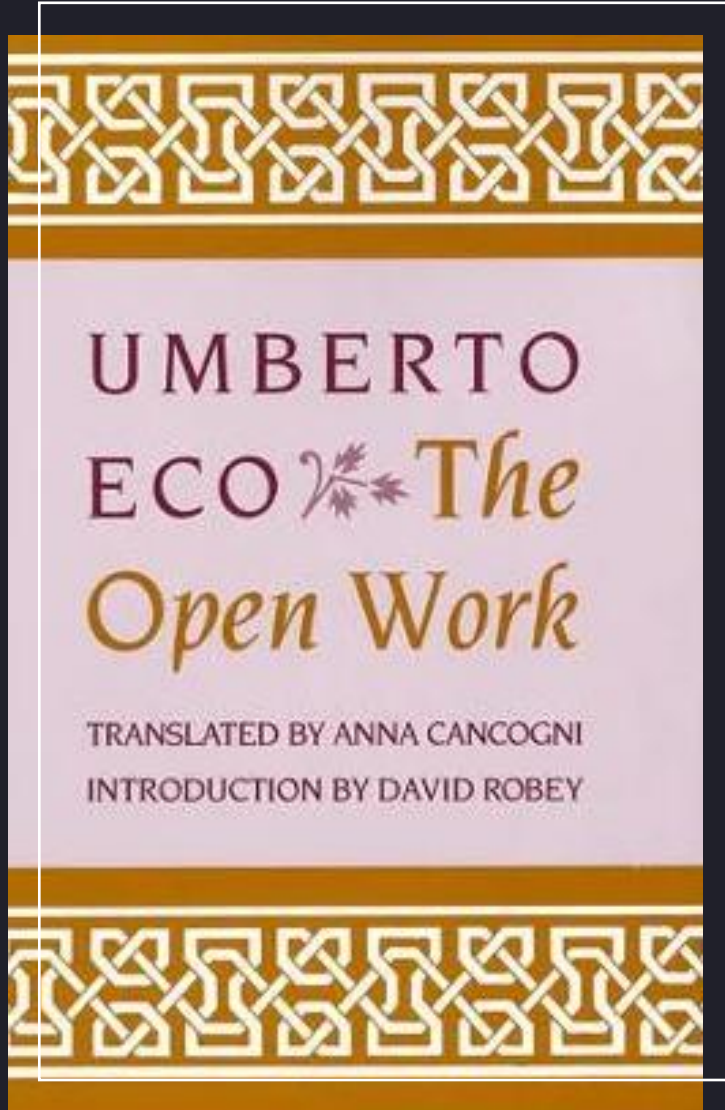


Lygia Clark, Bichos

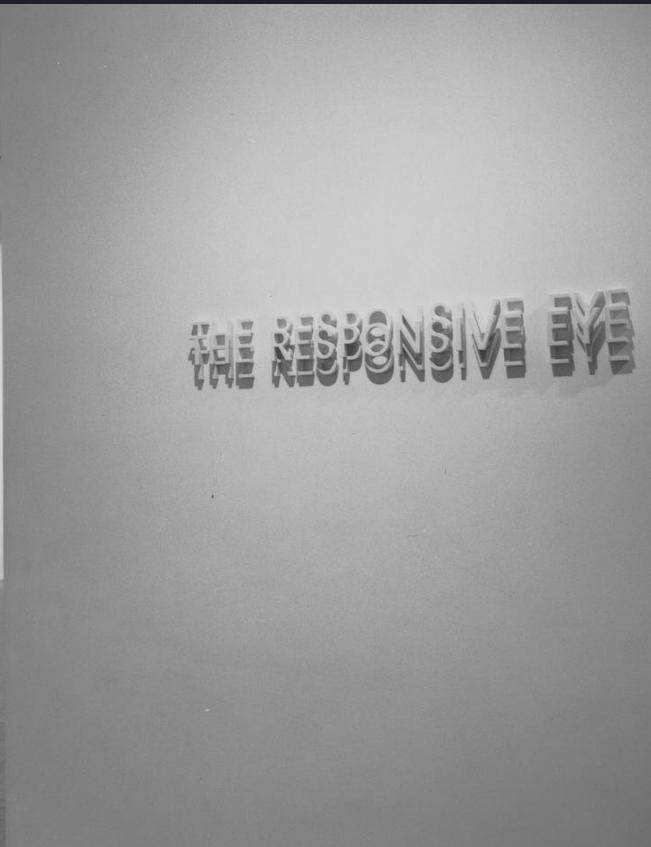
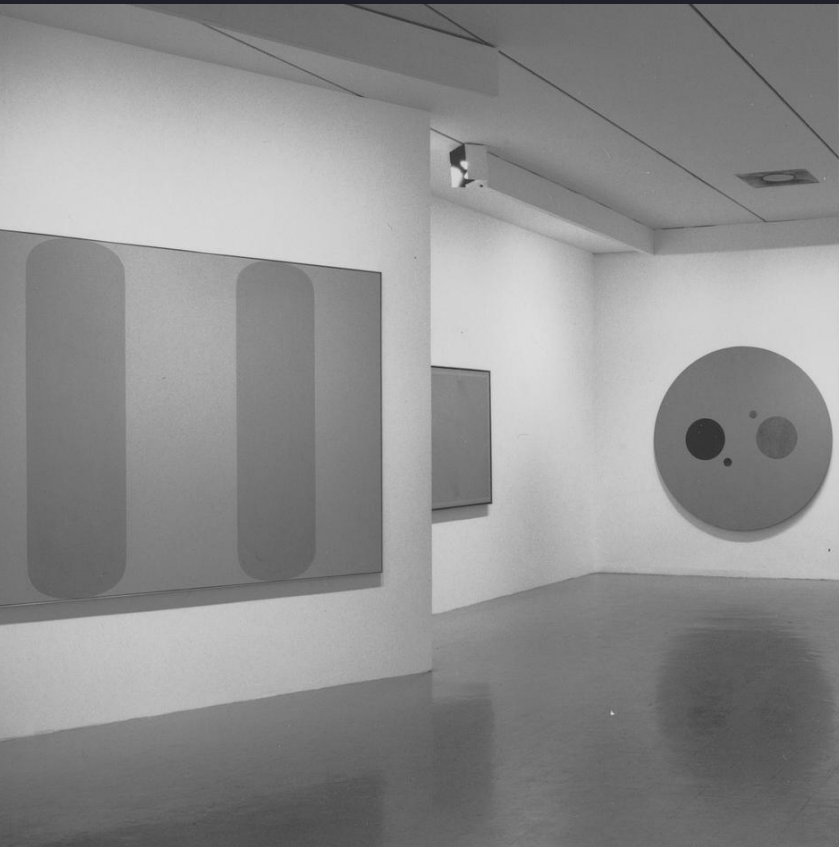


Roy Ascott

In his “metamechanics,” the machine does not gently respond to presence. The viewer must adapt to the system’s behavior.

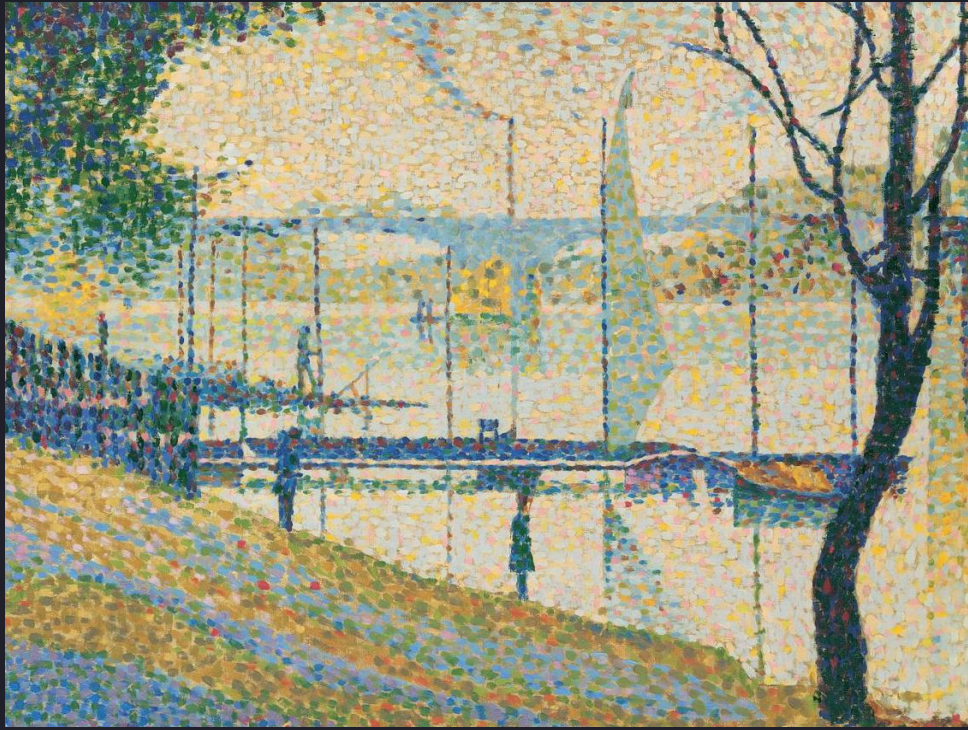


The artwork is a field of possibilities activated by the interpreter



1965, a watershed year for Kinetic and Op Art, marked by "The Responsive Eye" exhibition at New York's MoMA.

Op Art reveals interactivity as perceptual activation:
the retina and brain complete the work.

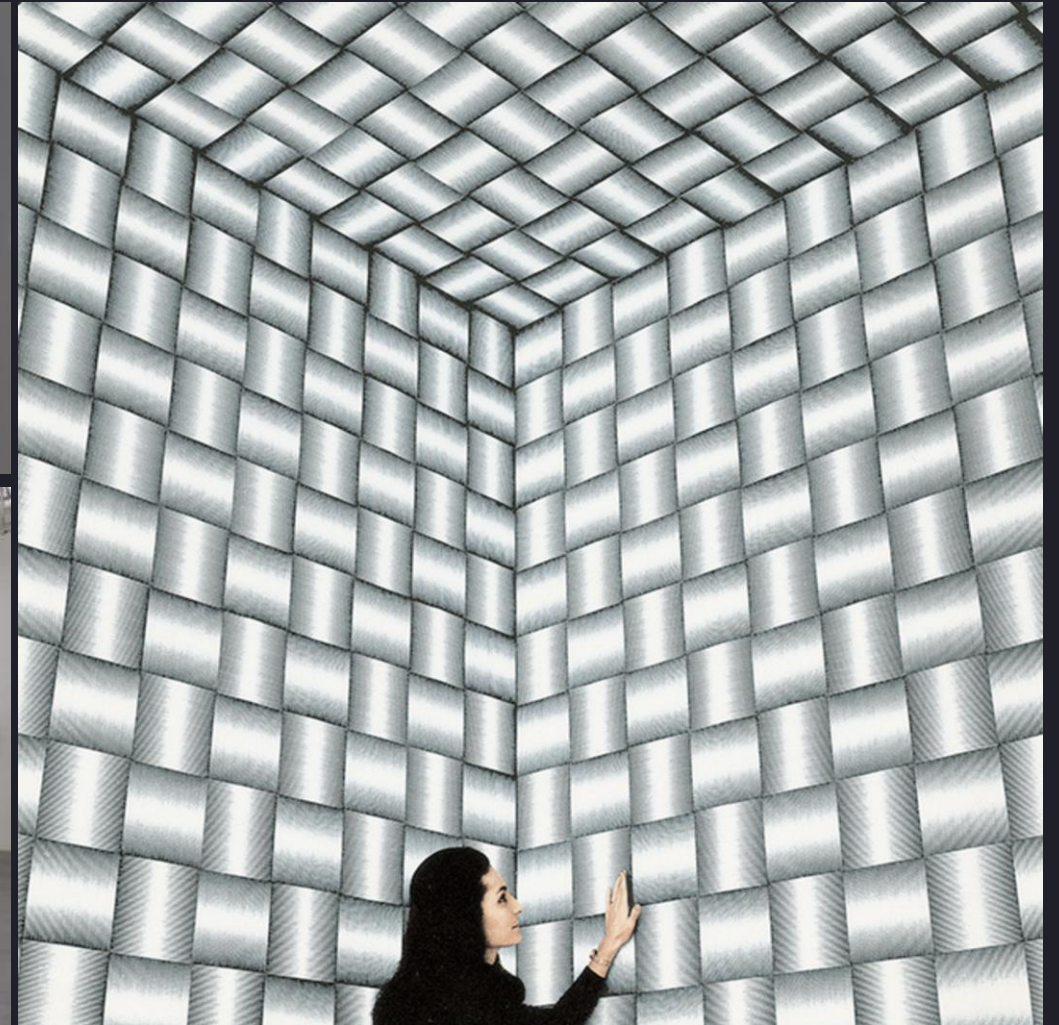


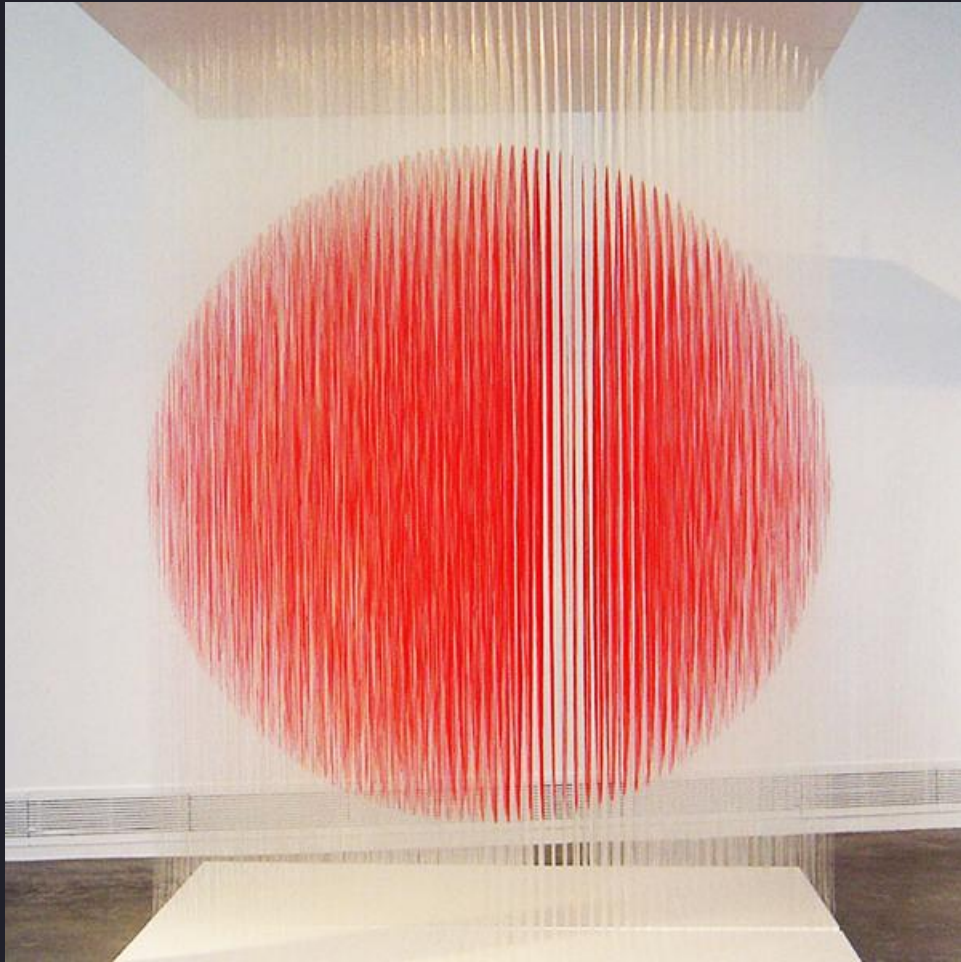
Georges Seurat



Bridget Riley

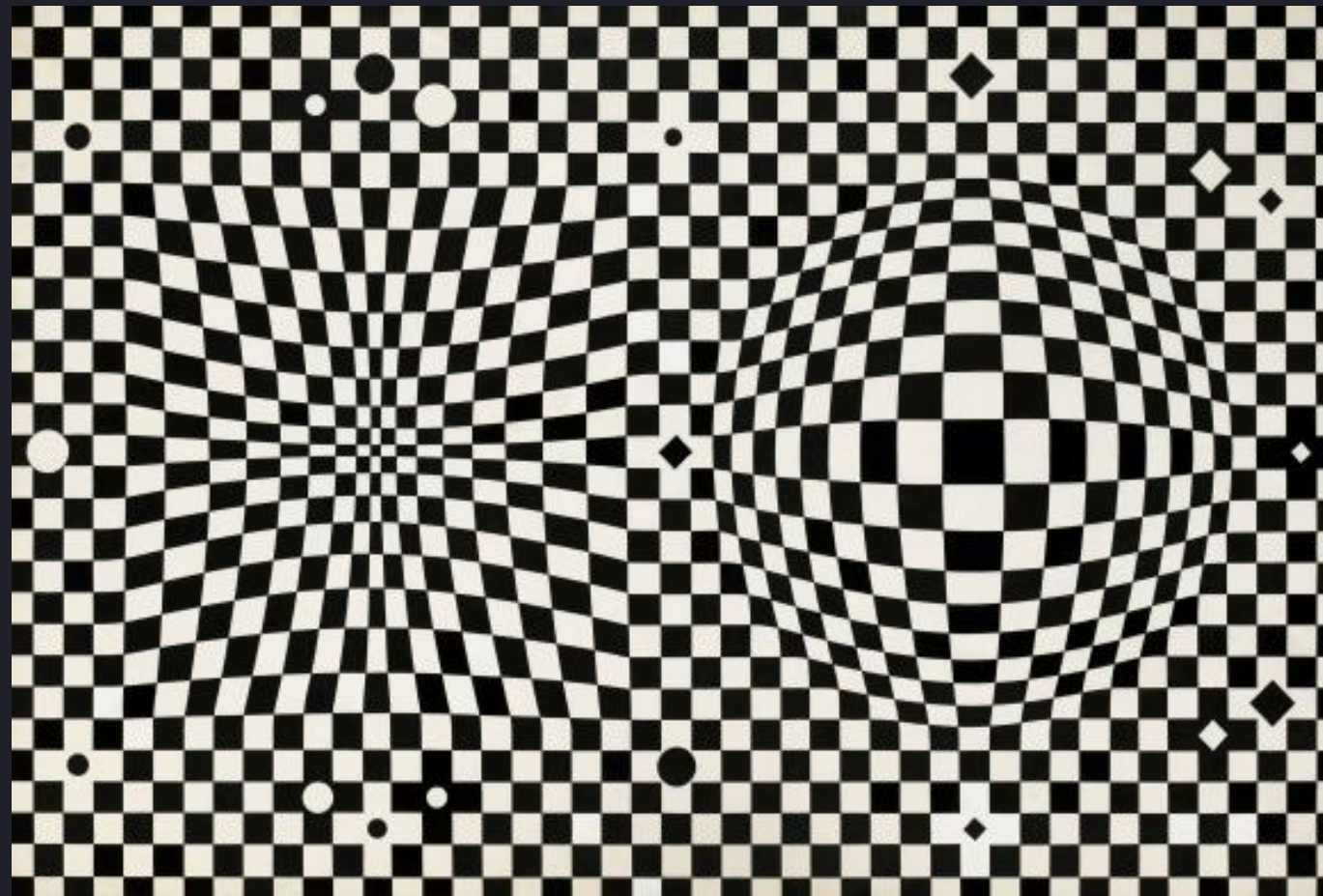
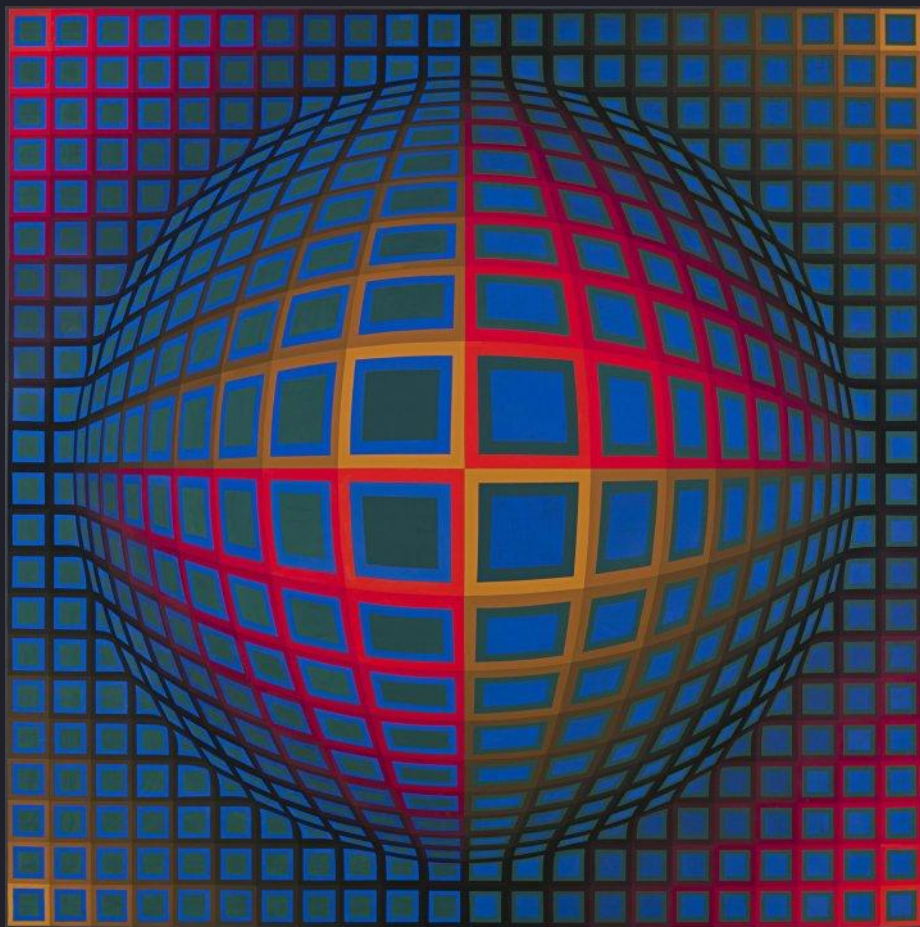
Getulio Alviani,
Vibrating Texture
Surfaces





Jesús Rafael Soto





Victor Vasarely's Vega



Victor Vasarely's Cité polychrome du bonheur/ Colorful city (1983)

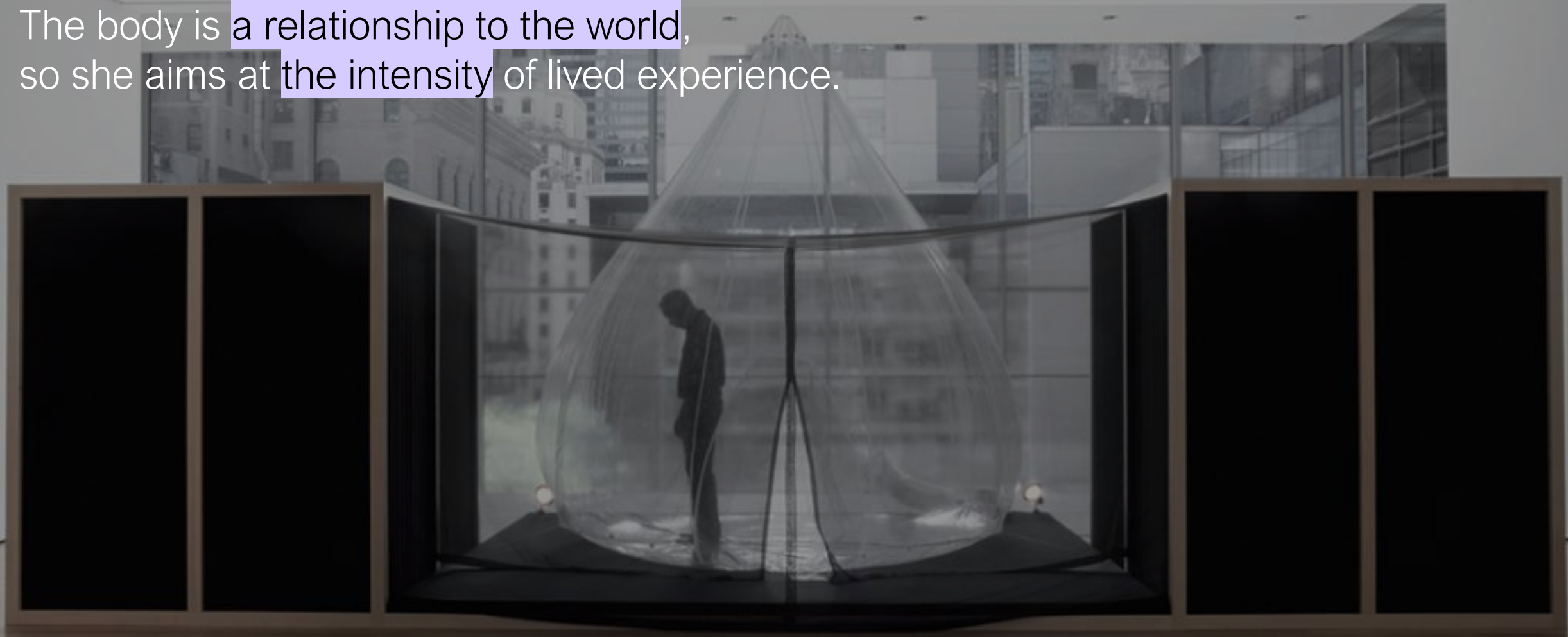
Art is meant to become an
instrument of social
engineering



I am not in front of my body, I
am in it or rather I am it.

Maurice Merleau-Ponty in *Phenomenology
of Perception*

The body is a relationship to the world,
so she aims at the intensity of lived experience.



Lygia Clark, The House Is the Body



The viewer becomes implicated in the culture of the favela.
This is an encounter with another social reality.

Hélio Oiticica, Parangolés



I am not in front of my body, Art becomes a field of encounter, a space of “being-together”

Nicolas Bourriaud

Inchiesta su "Modulo 856"

progettato da Enzo Mari in occasione della VI Biennale di San Marino.
(Scheda a cura di Umberto Eco)

Si prega di sottolineare la risposta con cui si concorda in ciascuna delle 3 colonne. Le conclusioni dell'inchiesta verranno pubblicate: il materiale potrà essere consultato dagli interessati. Le schede possono anche essere firmate.

Cos'è questo oggetto?

Vi piace?

- | | |
|---|------------------|
| a) è un'opera d'arte. | x) mi piace. |
| b) non è un'opera d'arte. | y) non mi piace. |
| c) è una macchina per
esperimenti psicologici. | z) mi irrita |

Cosa significa?

- 1) è il punto massimo della pop art: a furia di isolare oggetti reali ed esporli come opera, ora il visitatore è indotto a considerare se stesso, a mezzo busto, come quadro.
- 2) è il punto massimo della op art: ora lo spettatore è totalmente libero di determinare i movimenti dell'opera che guarda, e con l'appoggio della più classica delle illusioni ottiche, quella data dall'immagine speculare.
- 3) è la più completa fusione di op e di pop: viene esposto un oggetto reale, il visitatore, dotato del massimo di mobilità.
- 4) è un ritorno totale alla figurazione.
- 5) è una satira delle esigenze neofigurative.
- 6) è un invito a riconsiderare l'uomo.
- 7) irride, prendendoli in parola, a coloro che si lamentano che dall'arte contemporanea è assente l'uomo.
- 8) ripropone il mito di Narciso.
- 9) ripropone il motto del tempio di Delfi: "conosci te stesso".
- 10) vuole provocare il visitatore riconducendolo a se stesso, dopo la "distrazione" in cui si è trovato di fronte alle altre opere.
- 11) è un atto di protesta contro le mostre d'arte.
- 12) vuole mettere a disagio lo spettatore.
- 13) è il "divertissement" di chi accetta le mostre d'arte così come sono.
- 14) altra risposta a scelta:

Enzo Mari, Modulo 856

Would you like to ask your viewer something about your artwork? What would that be?

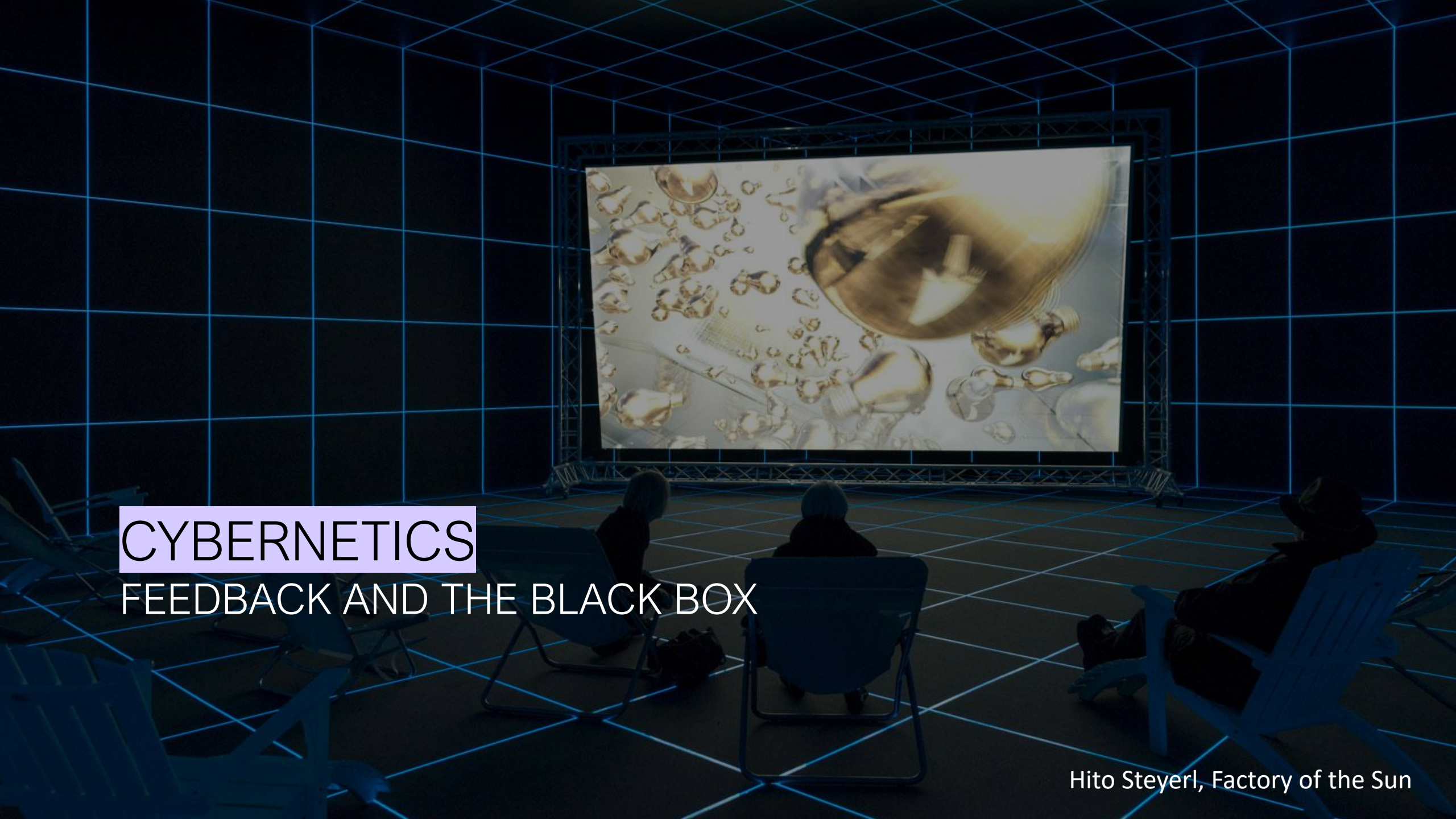
Is there a situation in your practice where you could creatively use questionnaires as part of the work?

Studio Reflections



KEY TAKEAWAYS

- Art shifts from static object to dynamic field. The artwork is no longer something to look at — it becomes something that unfolds through presence, movement, and time.
- The spectator moves from external observer to active component within the work. Change your position, your gesture, your attention — and the work changes.
- Interactivity emerges as a democratic gesture: activation replaces passivity, embodiment replaces distance, and art becomes a training ground for agency.
- By the late 1960s, participation is no longer only experienced — it is analyzed. The viewer's response becomes feedback, and feedback becomes data.



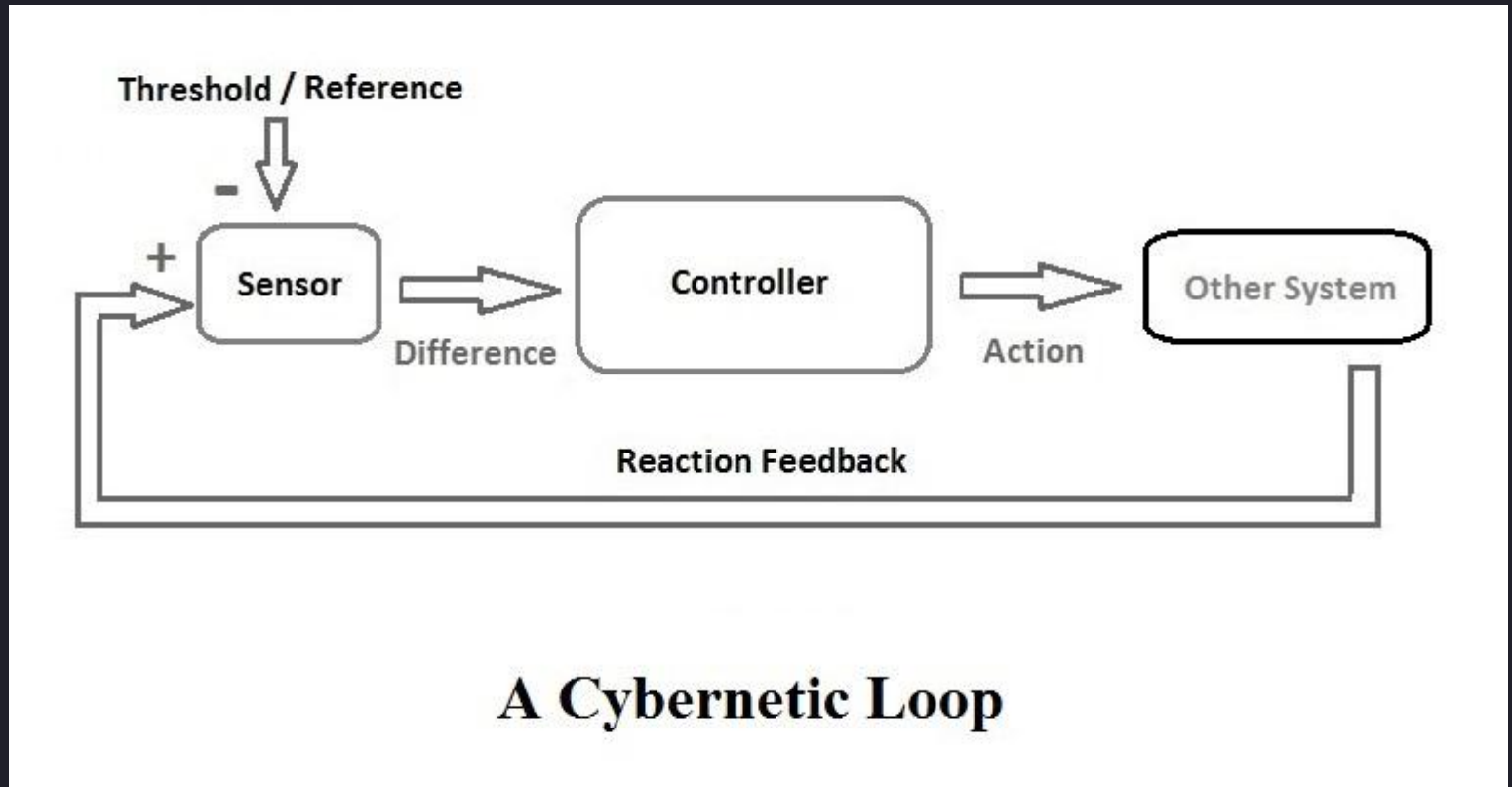
CYBERNETICS FEEDBACK AND THE BLACK BOX

Hito Steyerl, Factory of the Sun

If kinetic art questioned the stability of form,

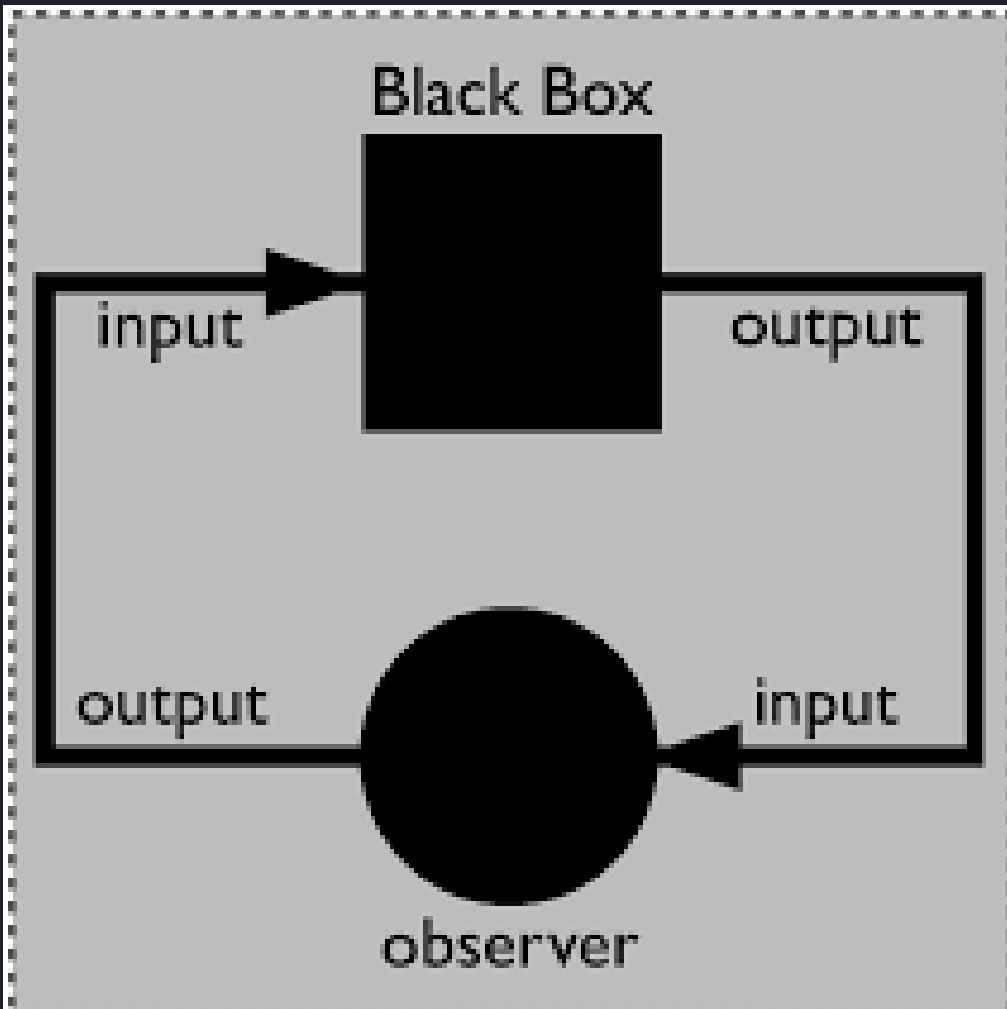
Op Art questioned the stability of perception,

cybernetics questioned the transparency of control.



A Cybernetic Loop

The system maintains itself in dynamic balance through a continuous exchange of signals, constantly correcting and recalibrating its actions in response to changing conditions



The spectator no longer sees how decisions are made.

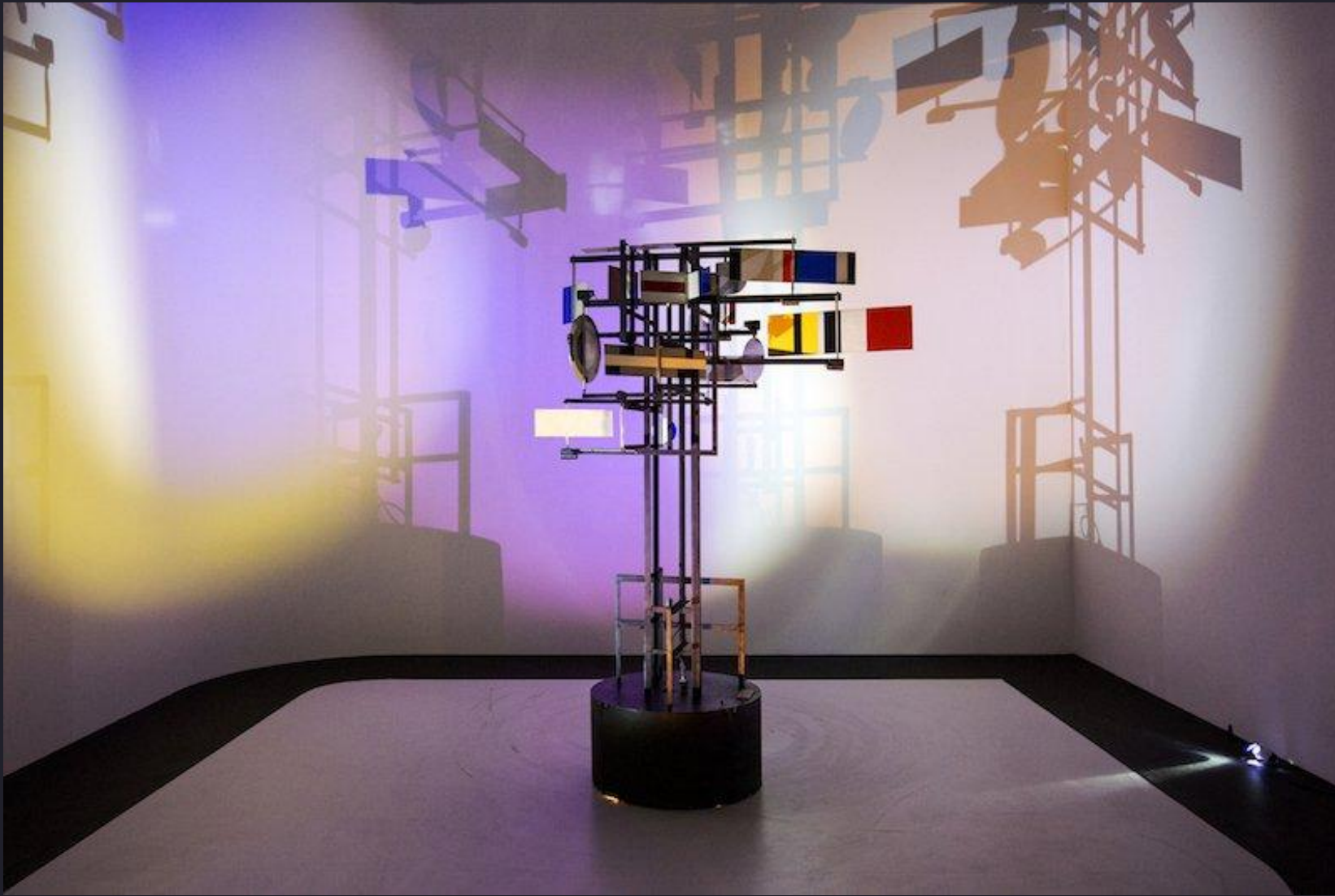
They no longer simply activate movement —their actions become part of a feedback loop.

CONFLICT 1 COMMUNICATION VS. CONTROL

WHO CONTROLS THE SYSTEM?

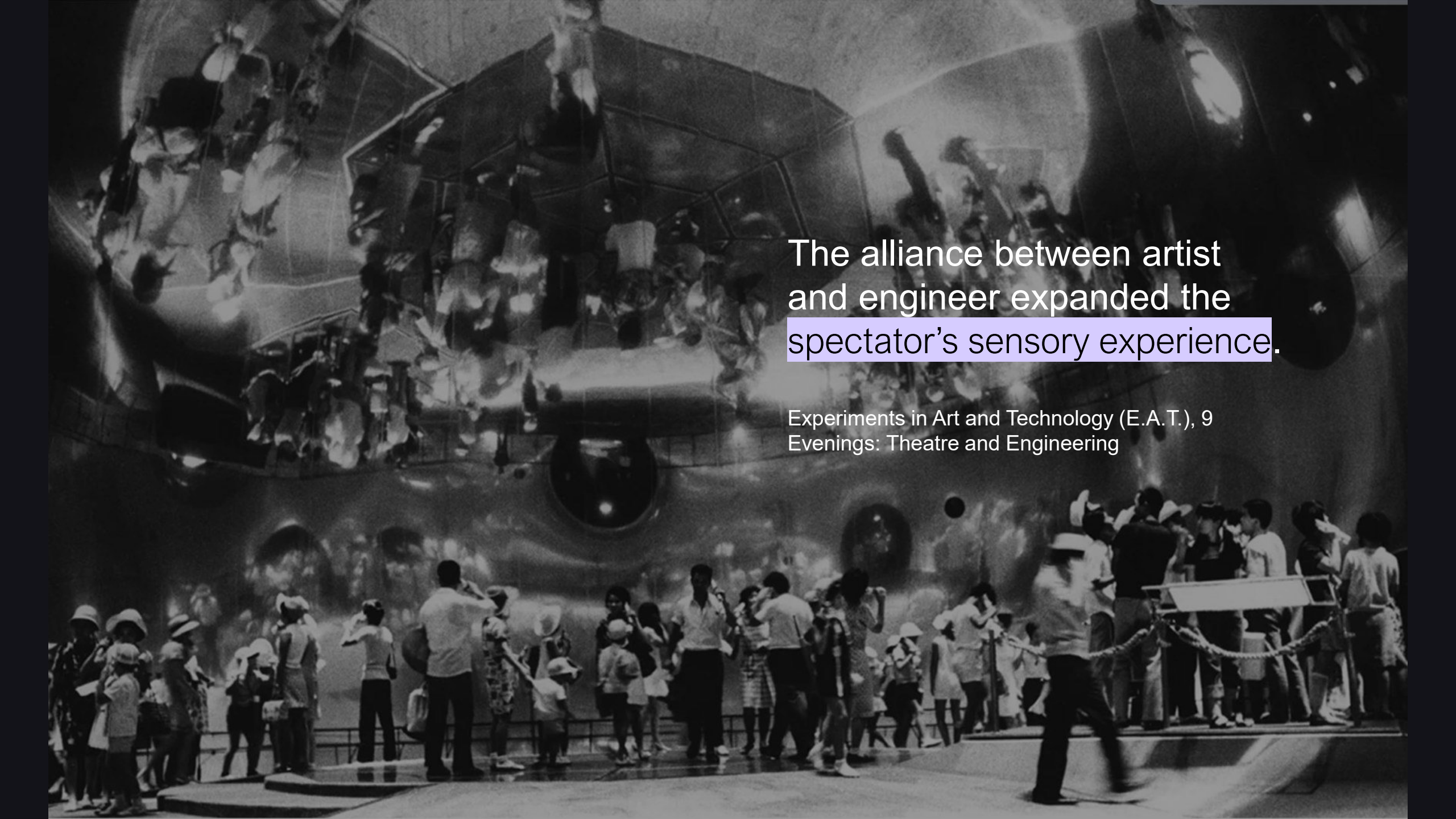


Gordon Pask,
Music Colour



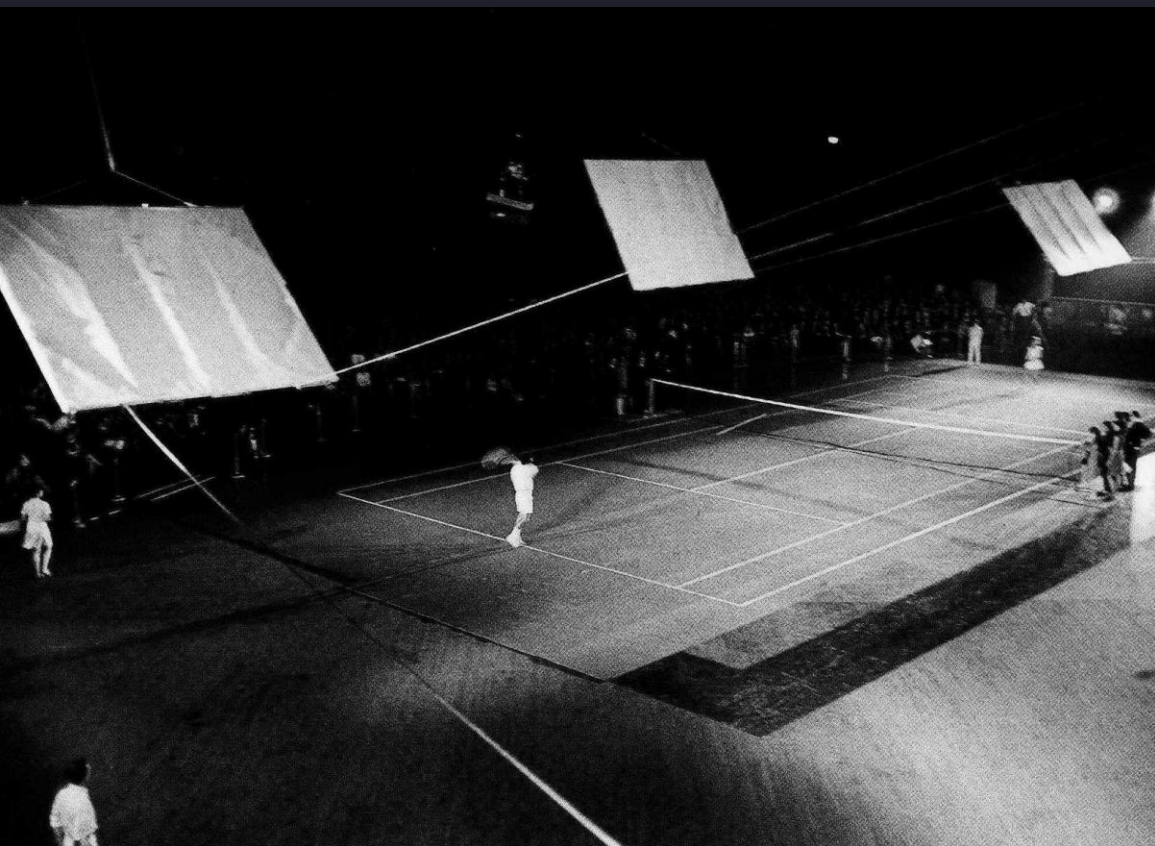
Cybernetic systems appear to democratize control by distributing it within networks of feedback, they also open the possibility of pervasive regulation.

Nicolas Schöffer, CYSP 1

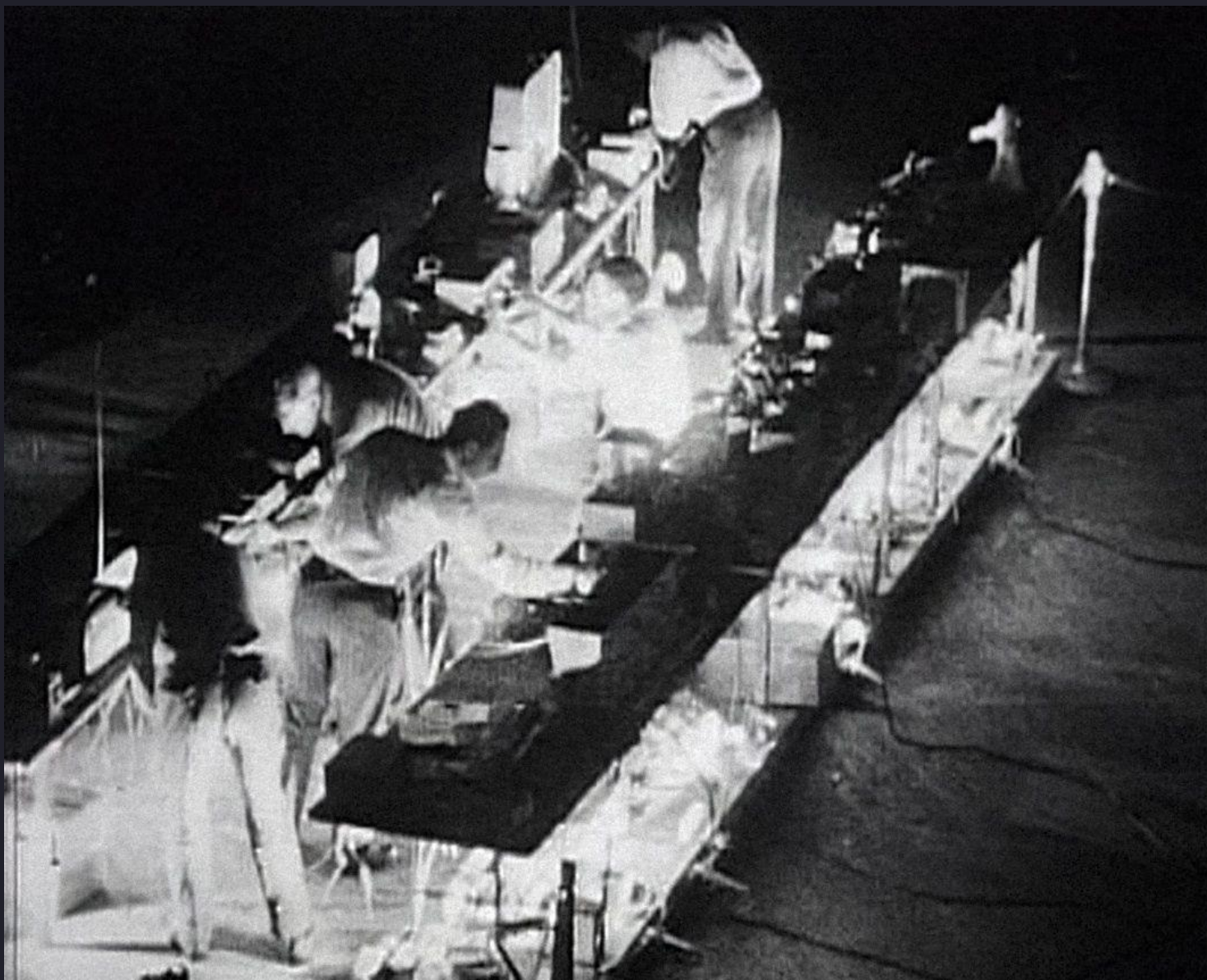


The alliance between artist
and engineer expanded the
spectator's sensory experience.

Experiments in Art and Technology (E.A.T.), 9
Evenings: Theatre and Engineering



Robert Rauschenberg, Open Score



Interactivity
disciplines body.

John Cage & David Tudor, Variations VII



Modern technology does not serve human purposes; it structures how the world appears to us. Everything becomes Bestand/ standing reserve.

Martin Heidegger

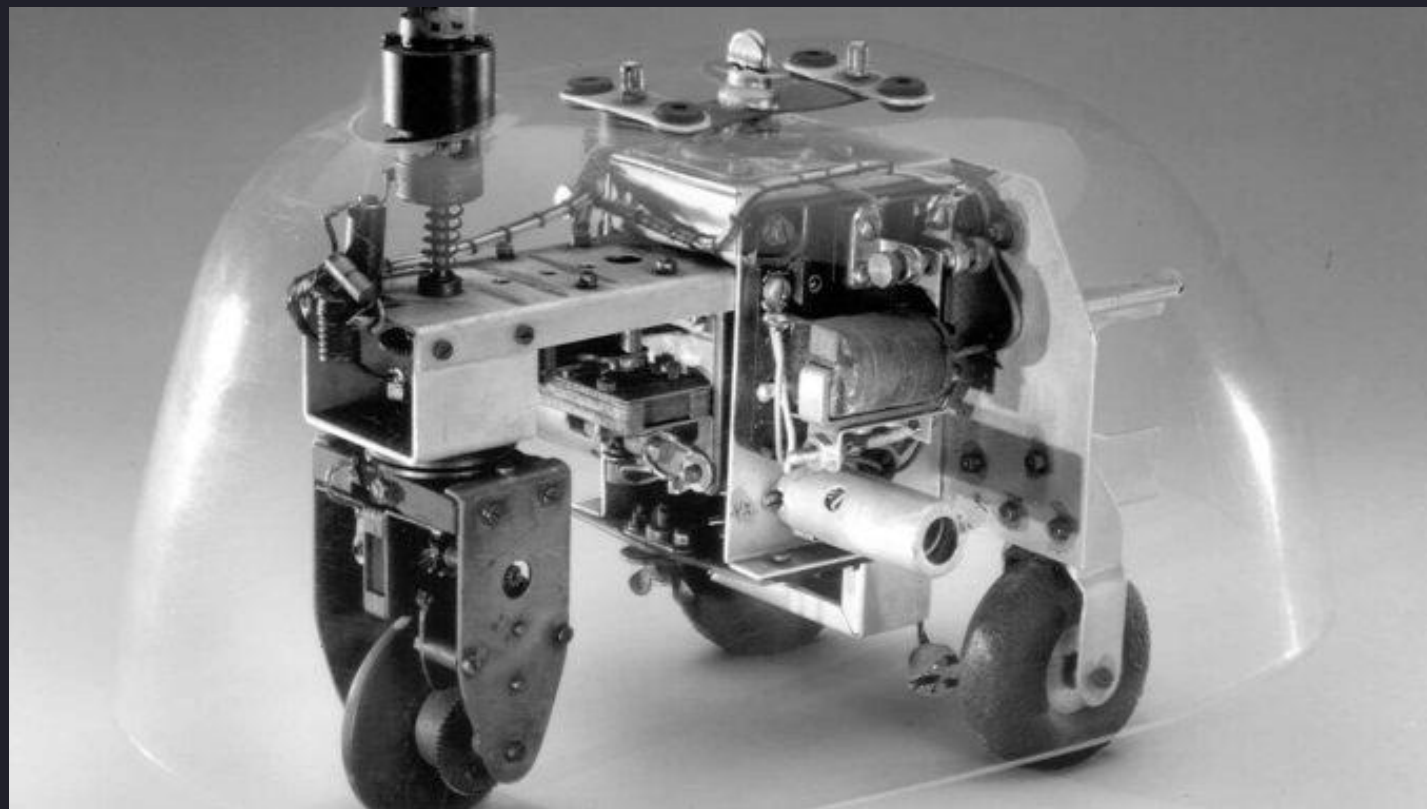
Interactivity enframes action before it occurs.

CONFLICT 2 PARTICIPATION VS. CONSTRAINTS

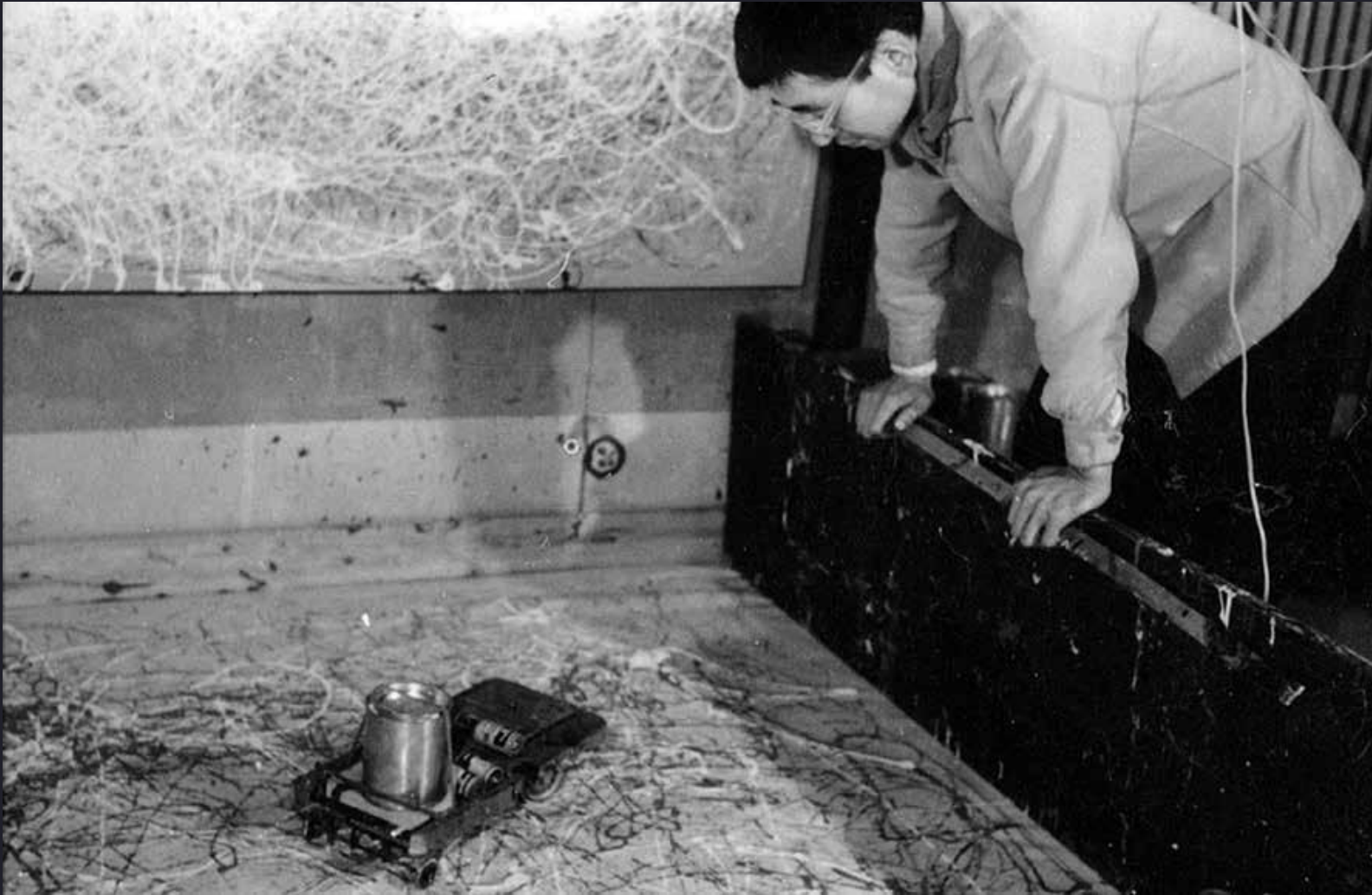
HOW MUCH CAN I ACTUALLY DO?

Conflict #1: The system listens and responds. But every response is already a correction. Here, interactivity is about who controls the feedback loop. Who controls the system?

Conflict #2 operates at the level of user experience and agency. The issue is no longer who governs the overall architecture, but what range of action is actually available within it.



W. Grey Walter, "tortoises" Elmer and CORA (1948–1952)

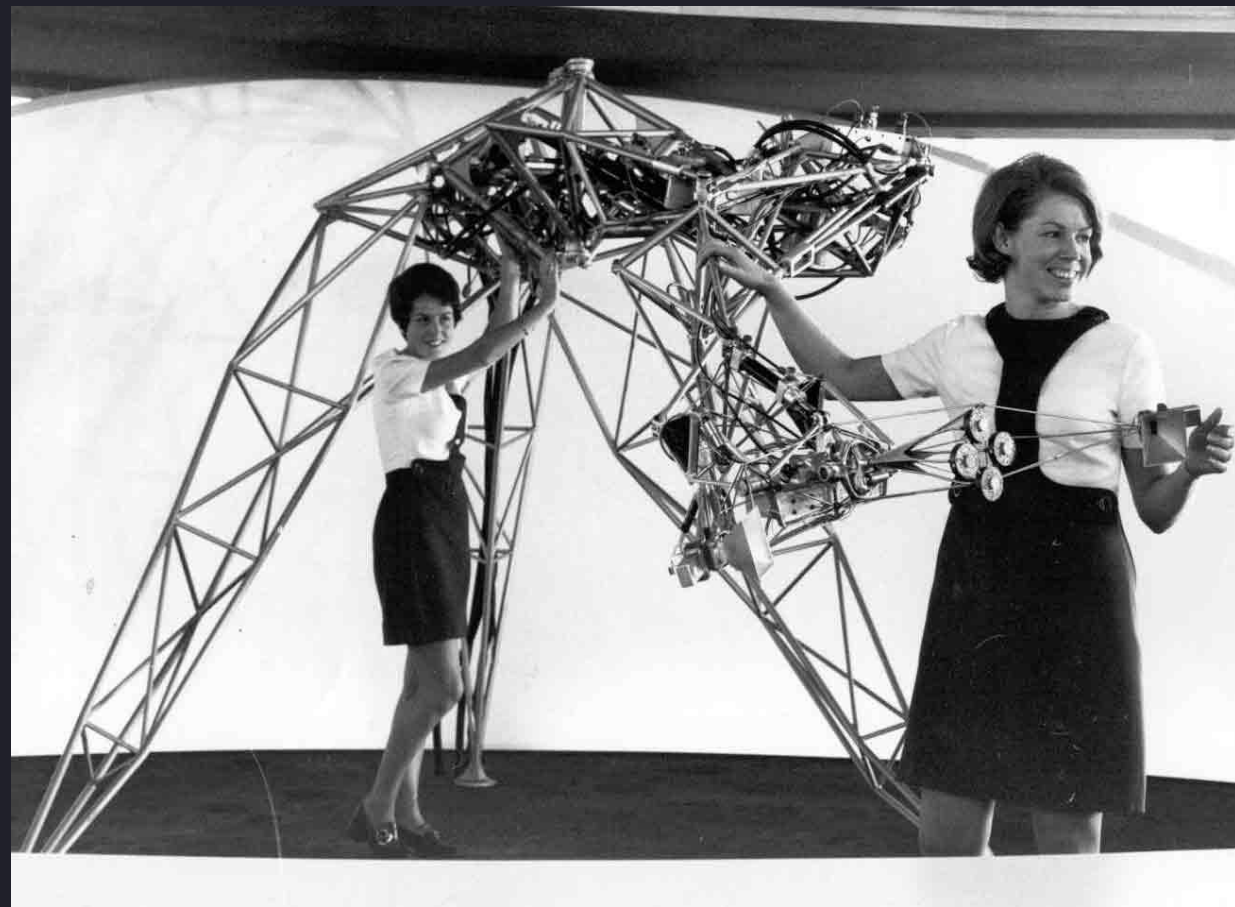
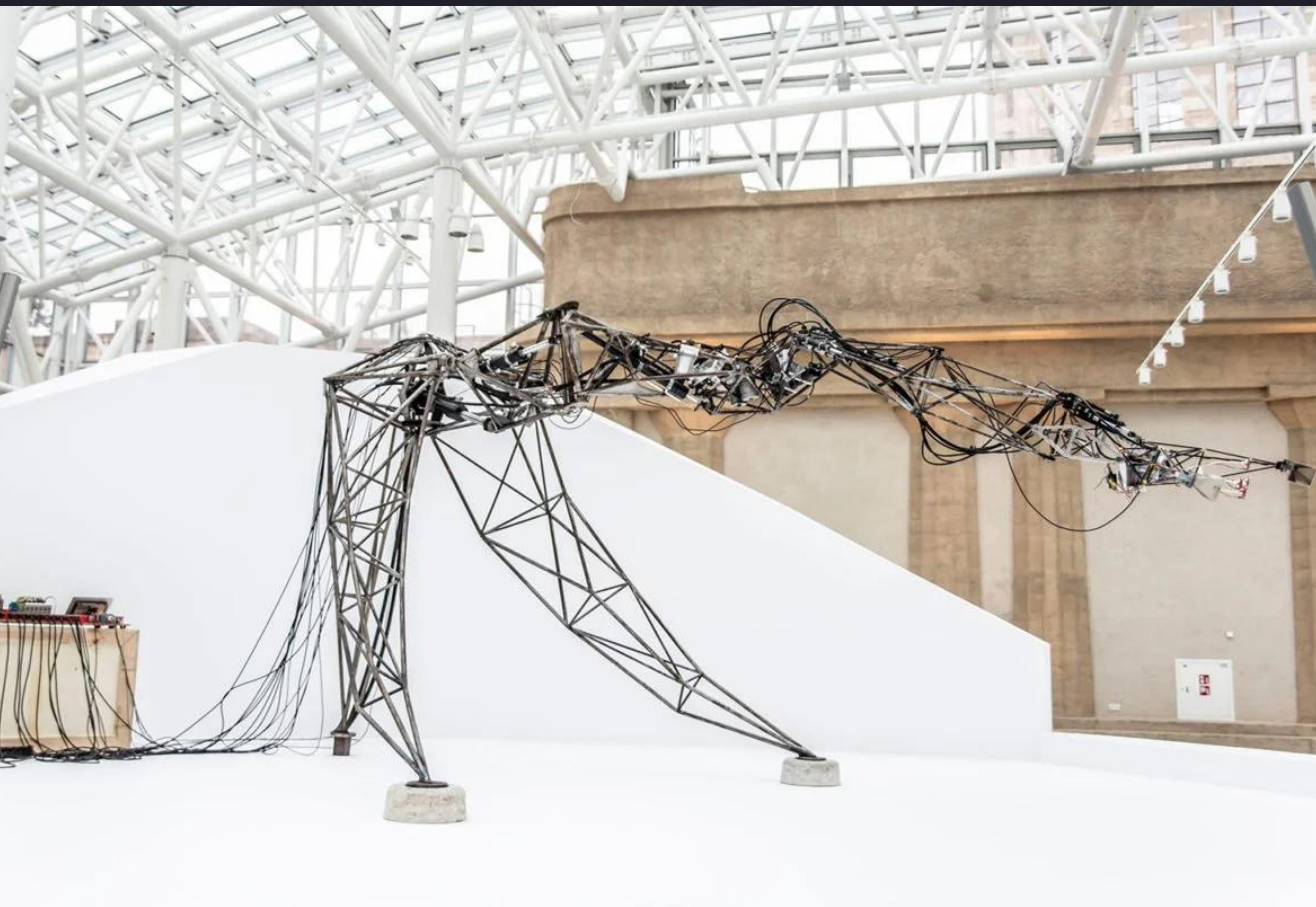


I am too busy with
Gutai secretarial
work, let the
machine paint.

Akira Kanayama, remote-controlled painting Gutai



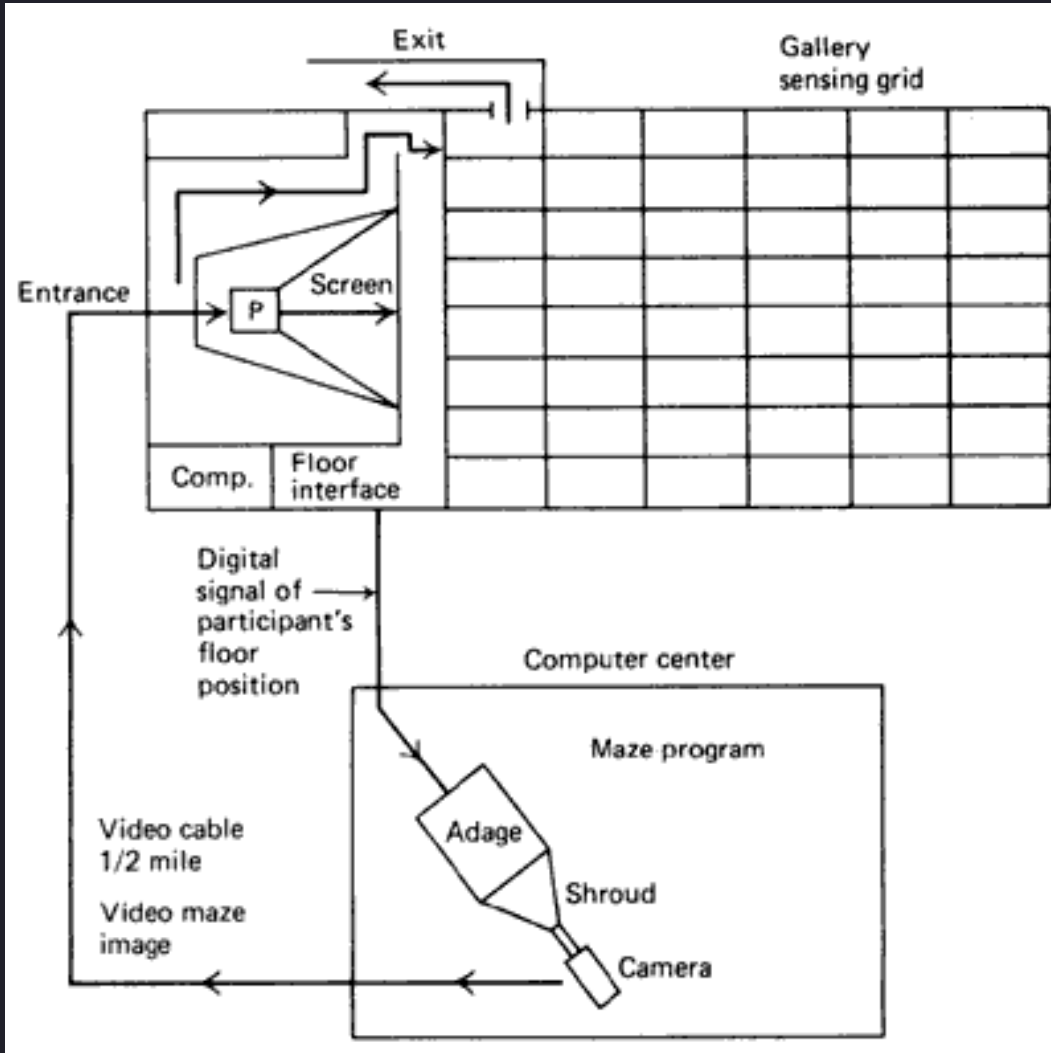
Edward Ihnatowicz, SAM (Sound Activated Mobile)



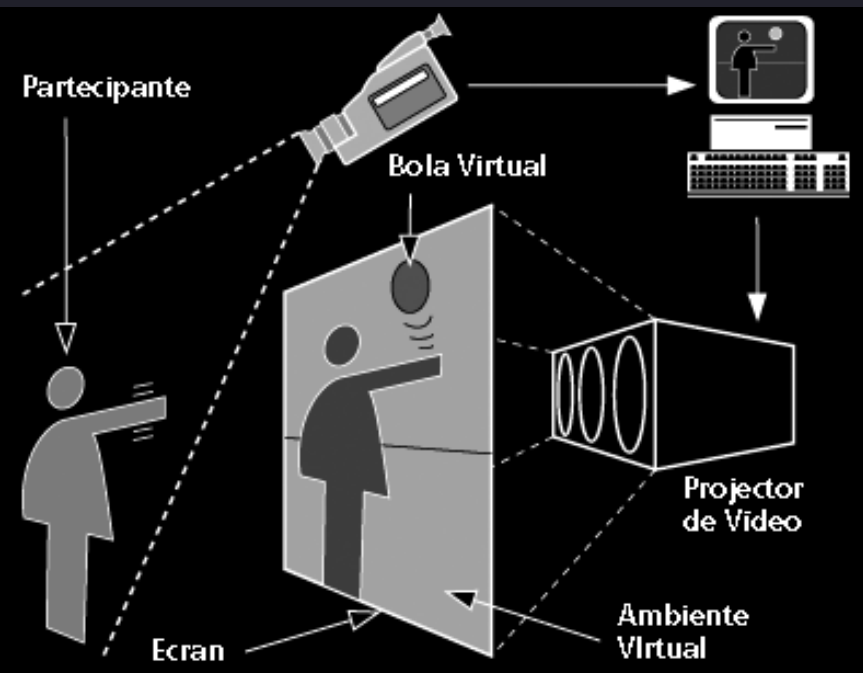
Edward Ihnatowicz, Senster



Gordon Pask, Colloquy of Mobiles



Myron Krueger, *Psychic Space* (1971)



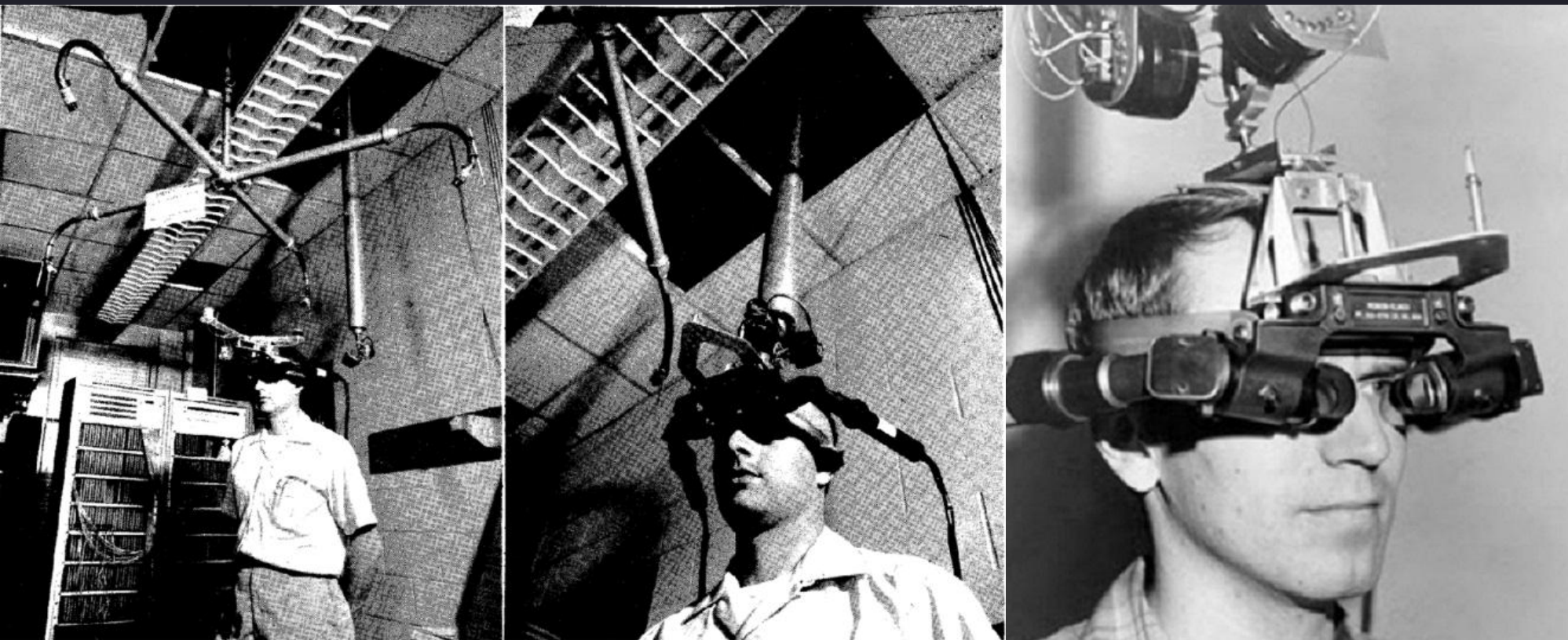
Myron Krueger, Video Place (1975)

The “smarter” the system appears, the easier it becomes to confuse programmed agency with freedom.

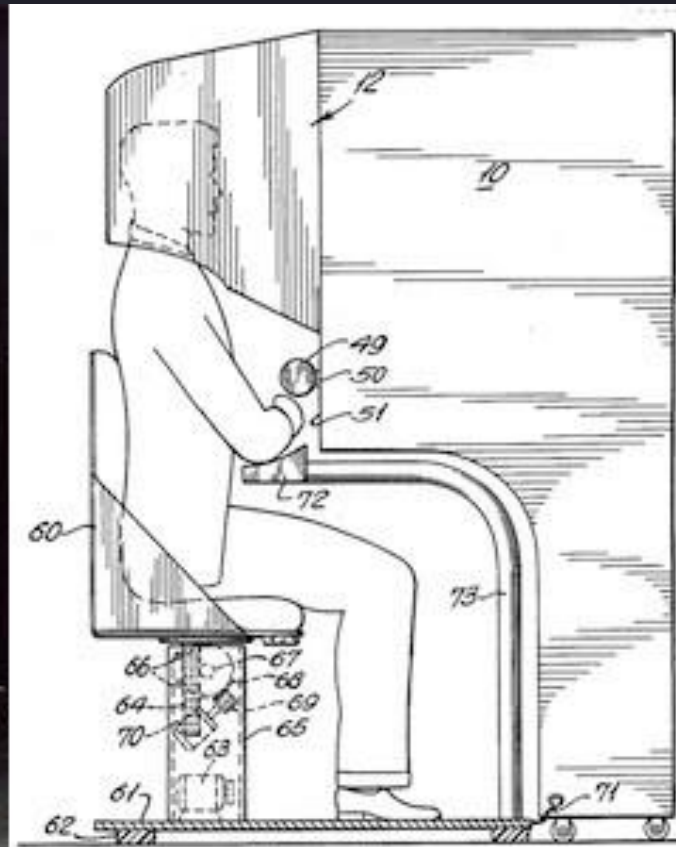
Cybernetic interactivity simultaneously trains subjectivity and exposes its limits .

CONFLICT 3 TRANSPARENCY VS. THE BLACK BOX

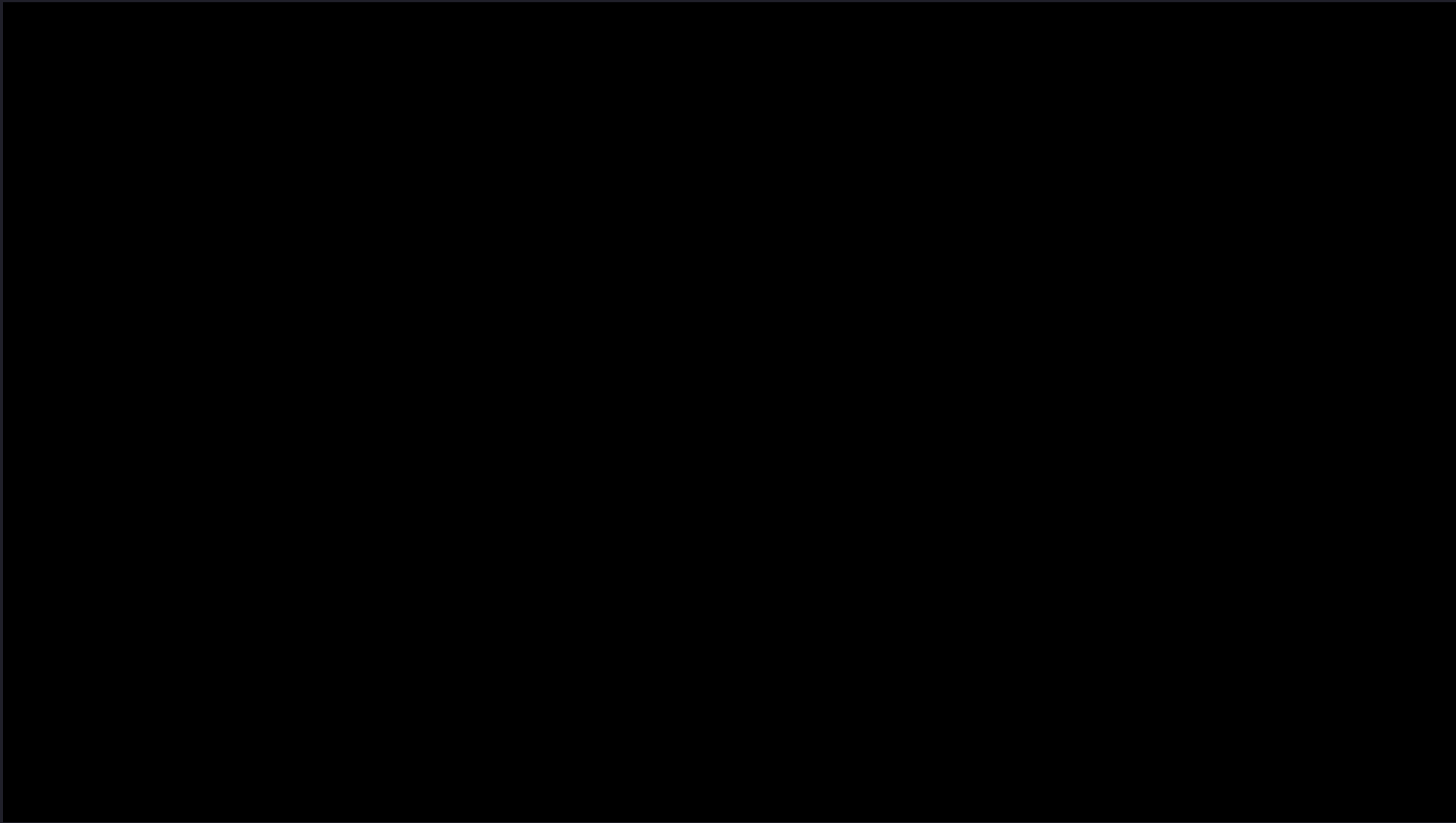
DO I UNDERSTAND WHAT IS HAPPENING?



Ivan Sutherland, The Sword of Damocles (1968)



Morton Heilig, Sensorama (1957)



David Rokeby, *Very Nervous System* (1986–90)

KEY TAKEAWAYS

- Cybernetics transforms interactivity from visible movement into invisible processing. The artwork no longer simply reacts, it adapts. Interactivity becomes a loop rather than a trigger.
- Every response is already a correction. The spectator appears to participate freely, but participation unfolds within an architecture designed to normalize behavior.
- Cybernetic artworks grant activity while defining its limits. The viewer acts and adapts, yet the range of responses is pre-coded. Freedom exists within rules; interactivity becomes the design of behavior.
- As systems grow more complex, their logic becomes opaque. We see input and output, but not the process in between. The viewer no longer stands before the work, but inside its feedback loop. Interactivity becomes procedural and increasingly invisible.